

# The Primitive Generalized Fermat Equation

$$x^3 + y^5 = z^7$$

A computer-assisted proof

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## Abstract

We prove that the generalized Fermat equation

$$x^3 + y^5 = z^7$$

has no solution in nonzero coprime integers, and we make explicit how the proof extends the prior work cited below. Dahmen–Siksek had established the signed local descent and eliminated the three cases in which the associated degree-seven algebra is reducible. Putz had then proved that the remaining irreducible case can involve only seven septic fields: six pure fields and one exceptional field. The missing step was to exclude those seven fields.

For the six pure fields we construct an explicit Fano resolvent, descend to a smooth plane quartic of genus three over  $\mathbf{Q}(\sqrt{-7})$ , and prove that its rational-parameter locus consists only of five points above the branch values. For the exceptional field we combine the modularity and conductor results of Pacetti–Villagra Torcomian with level lowering over  $\mathbf{Q}(\sqrt{5})$  and a finite Hecke comparison at 29. We also reconstruct the three reducible-sector arguments of Dahmen–Siksek as independently replayable calculations, adding a database-free identification of the quadratic field and an intrinsic formal-group treatment at the ramified prime. The paper labels every major step as a literature input, reconstructed input, or new argument. All project-specific finite calculations are supplied as exact certificates, programs, inputs, and authenticated logs.

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# 1 Introduction: the frontier and the completion

## 1.1 The equation and the theorem

For pairwise coprime nonzero integers, the equation

$$x^3 + y^5 = z^7 \tag{1}$$

is the generalized Fermat equation of signature  $(3, 5, 7)$ . We shall use the variable order

$$(X, Y, Z) = (y, x, z), \quad X^5 + Y^3 = Z^7. \tag{2}$$

All signs are allowed. Since a prime dividing two of  $X, Y, Z$  also divides the third, primitivity is equivalent here to pairwise coprimality. Our main result is the following.

**Theorem 1.1.** *There are no nonzero primitive integers  $x, y, z$  satisfying*

$$x^3 + y^5 = z^7.$$

The purpose of this introduction is not only to state the theorem, but to say precisely what was known before the present work and what had still to be done. This distinction matters in a long computer-assisted proof: it makes clear which general theorems are imported, which earlier computations are reconstructed for confidence, and which arguments close the remaining gap.

## 1.2 Prior work

The mathematical starting point was already far advanced. Dahmen and Siksek proved the following strong partial result [1, Theorem 1]: every nontrivial primitive solution of  $X^5 + Y^3 = Z^7$  satisfies one of the alternatives

$$2 \cdot 3 \cdot 5 \mid Z \text{ and } 7 \nmid XY, \quad \text{or} \quad 2 \nmid Z, 3 \nmid XYZ, 5 \nmid YZ, 7 \mid Y. \quad (3)$$

In particular, a primitive solution for which  $3 \mid XY$ ,  $5 \mid Y$ , or  $7 \mid X$  is trivial. Their descent introduces the degree-seven algebra used below, proves that only four factor-degree patterns can occur, and eliminates the three reducible patterns

$$[1, 6], \quad [2, 5], \quad [3, 4].$$

The only surviving pattern is therefore the irreducible one, [7].

Putz then made this residual problem finite. His exhaustive Hunter enumeration, with the prescribed local algebra data, proves that an irreducible descent algebra must be one of exactly seven fields [2, Theorem 3.50]. Six are pure septic fields and the seventh is an exceptional septic field. Thus the established frontier can be summarized in one sentence:

*a nontrivial primitive solution would have to pass through one of seven explicit irreducible degree-seven fields.*

This is a remarkably narrow frontier, but it is not yet a contradiction. One must still connect each field to an arithmetic object on which rational points or residual Galois representations can be exhausted.

## 1.3 What is added here

The principal new work is the elimination of those seven fields.

- (i) For the six pure fields we construct the missing geometric bridge. The thirty Fano planes attached to the septic cover split into two  $A_7$ -orbits. Their compatible pairs give a degree-120 resolvent whose quotient is an explicit smooth plane quartic  $C_+$  over  $\mathbf{Q}(\sqrt{-7})$ . A rational-parameter theorem shows that the relevant parameter on  $C_+$  takes rational values only at five points above 0, 1, and  $\infty$ . A nonzero primitive solution would have parameter outside that set, so all six pure fields are excluded.
- (ii) For the exceptional field we place the specialization in the Frey family of Pacetti–Villagra Torcomian, use their modularity and conductor theorems together with Hilbert level lowering, and reduce both the irreducible and reducible residual alternatives to finite lists. At a prime above 29, the four curve trace polynomials and the six possible modular or character factors have 24 nonzero resultants modulo 7. This excludes the exceptional field.
- (iii) We reconstruct the reducible-sector arguments of Dahmen–Siksek in an exact certificate framework. This is not presented as a new elimination of those sectors. It provides an independent route through the finite arithmetic and adds two genuine refinements: the quadratic factor is identified from its intrinsic local labels without a field database, and the ramified 7-adic step is carried out inside the formal neighbourhood of the given integral Jacobian model.

- (iv) We compose all four sectors into a single machine-checkable theorem graph. Every project-specific finite assertion is tied to exact input bytes, an explicit predicate, and a replay program or authenticated completed log. General theorems from the literature remain ordinary bibliographic dependencies.

#### 1.4 A provenance-aware proof map

The following table is the shortest guide to the logical ownership of the proof. “Reconstructed” means that this paper supplies an independent exact certificate for a result whose mathematical argument was already in the literature; it does not claim priority for that result.

| stage                                        | status before this paper                            | role in this paper                                                               |
|----------------------------------------------|-----------------------------------------------------|----------------------------------------------------------------------------------|
| signed local descent and four factor sectors | Dahmen–Siksek                                       | imported, restated, and independently checked                                    |
| sectors [1, 6], [2, 5], [3, 4]               | eliminated by Dahmen–Siksek                         | reconstructed with certificates; two local refinements are added                 |
| irreducible sector [7]                       | reduced by Putz to seven explicit fields            | the seven-field list is imported and authenticated                               |
| six pure septic fields                       | not eliminated by the cited prior work              | excluded here by the Fano resolvent and the rational-parameter theorem           |
| one exceptional septic field                 | modular tools available, but no terminal comparison | excluded here by level lowering, ray characters, and the $q = 29$ resultant test |
| global theorem                               | one finite irreducible frontier remained            | all four sectors are composed into the contradiction of Theorem 1.1              |

#### 1.5 How the proof was completed

The working principle was to treat a hypothetical solution not as a large integer triple to be chased, but as one mathematical state seen through several exact representations. The same state has a descent algebra, local factor labels, ideal classes, divisor classes, finite-quotient coordinates, and  $p$ -adic analytic coordinates. Each representation supplies a constraint. The proof succeeds when the intersection of all compatible states is empty.

This viewpoint led to the following order of work:

1. fix the strongest established frontier before beginning a new search;
2. translate each remaining assertion into a finite interface with exact inputs and a stated acceptance predicate;
3. use exploratory computation only to discover a model, prime, or quotient, and then replace discovery by a deterministic certificate;
4. keep the four descent sectors separate until each has a terminal statement, and only then compose them;
5. distinguish a recomputation of an existing argument from a genuinely new bridge or obstruction.

This is why the proof is modular. A specialist can inspect the new irreducible argument without repeating the reducible sectors, or can replay one finite computation without trusting the remainder of the software tree.

The central object throughout is the polynomial

$$\Phi(T) = 15T^7 - 35T^6 + 21T^5, \quad \eta = \frac{X^5}{Z^7}, \quad f_\eta(T) = \Phi(T) - \eta. \quad (4)$$

The associated étale algebra

$$A_\eta = \mathbf{Q}[T]/(f_\eta)$$

encodes a hypothetical solution in several complementary forms. Local factorization describes which global fields may occur. Descent maps those fields to curves. Mordell–Weil information restricts the resulting points to finitely many congruence classes, and local analytic functions decide whether those classes contain a point compatible with the original rational parameter.

The argument uses standard results in descent, number-field enumeration, modularity, level lowering, formal groups, Mordell–Weil sieves, and Chabauty methods. The large finite calculations specific to (1) are exact and are separated into independently checkable certificates. Thus the proof has two visible layers: general theorems, cited in the usual way, and finite arithmetic, supplied with the paper. Sections 2–5 reconstruct the descent and reducible sectors. Section 6 begins at Putz’s seven-field frontier and contains the new terminal argument. The final sections compose the proof and describe the certificates.

## 2 The degree-seven descent algebra

**Provenance.** The local covering and exhaustive four-sector reduction in this section are inputs from Dahmen–Siksek. We record them because they are the interface between a primitive solution and every later computation.

The derivative and the two branch values are given by

$$\Phi'(T) = 105T^4(T-1)^2, \quad \Phi(0) = 0, \quad \Phi(1) = 1. \quad (5)$$

In particular,  $\Phi$  is strictly increasing on  $\mathbb{R}$ . For a nonzero solution,  $\eta \notin \{0, 1\}$ , and  $f_\eta$  is separable with one real root. A resultant computation gives

$$\text{disc}(f_\eta) = -3^6 5^6 7^7 \eta^4 (\eta - 1)^2. \quad (6)$$

Its squareclass is therefore  $-7$ .

The local covering theorem at 7 in [1, Proposition 3.5] gives local factor-degree patterns  $[1, 6]$ ,  $[2, 5]$ ,  $[3, 4]$ , and  $[7]$ . Global factors induce groupings of the irreducible local factors, so their degrees are obtained by coarsening these patterns. Consequently the possible global factor-degree patterns are exactly among

$$[1, 6], \quad [2, 5], \quad [3, 4], \quad [7]. \quad (7)$$

In particular, there are at most two global factors. For example, if  $7 \mid X$  and  $r = \text{ord}_7(X)$ , the Newton polygon has vertices  $(0, 5r)$ ,  $(5, 1)$ ,  $(7, 0)$ , giving irreducible local factors of degrees 5 and 2. If  $7 \mid Y$  and  $r = \text{ord}_7(Y)$ , translation by 1 gives vertices  $(0, 3r)$ ,  $(3, 1)$ ,  $(7, 0)$ , giving irreducible local factors of degrees 3 and 4. The unit case, and the  $7 \mid Z$  case after scaling to an integral model, give the remaining local patterns by the cited proposition.

We now eliminate the four sectors one at a time.

## 3 The rational-factor sector

**Provenance.** The mathematical elimination in this section is due to Dahmen–Siksek. This section gives a certificate-level reconstruction of their Proposition 6.1 and Lemmas 6.2–6.8 [1, Section 6].

Assume that  $r \in \mathbf{Q}$  is a root of  $f_\eta$ , and put  $w = rZ$ . Clearing denominators gives

$$15w^7 - 35Zw^6 + 21Z^2w^5 - X^5 = 0. \quad (8)$$

The translated identity

$$\Phi(T) - 1 = (T - 1)^3(15T^4 + 10T^3 + 6T^2 + 3T + 1)$$

also gives

$$(w - Z)^3(Z^4 + 3Z^3w + 6Z^2w^2 + 10Zw^3 + 15w^4) = -Y^3. \quad (9)$$

### 3.1 Local restrictions and quadratic ideal descent

Newton polygons at 3, 5, and 7 give

$$3 \nmid Z, \quad 5 \nmid YZ, \quad 7 \nmid Y, \quad \text{ord}_3(r) \in \{0, 1, -1\}, \quad \text{ord}_5(r) \geq 0. \quad (10)$$

The rational-root theorem applied to (8) then shows that  $w$  is an integer or one third of an integer. Moreover, for every prime  $p$ , at least one of  $\text{ord}_p(Z)$  and  $\text{ord}_p(w)$  is zero.

For all three reducible sectors fix the notation

$$k = \mathbf{Q}(\omega) = \mathbf{Q}(\sqrt{-35}), \quad \omega^2 + \omega + 9 = 0, \quad \delta = 2\omega + 1 = \sqrt{-35}. \quad (11)$$

Thus  $\mathcal{O}_k = \mathbf{Z}[\omega]$ . Put

$$A = Z^2 + wZ - 3w^2, \quad B = -wZ - w^2, \quad \mathcal{F} = A + B\omega.$$

Then

$$\mathcal{F}\overline{\mathcal{F}} = A^2 - AB + 9B^2 = Z^4 + 3Z^3w + 6Z^2w^2 + 10Zw^3 + 15w^4.$$

The class group of  $k$  is  $C_2$ . The prime 3 splits, and exact ideal arithmetic reduces (9) to three possibilities:

$$\mathcal{F} = \alpha(u + v\omega)^3, \quad \alpha \in \{1, 9\omega, 3\omega^2\}, \quad u, v \in \mathbf{Q}. \quad (12)$$

For  $\alpha = 1$ , [1, Lemma 6.4] further gives  $w, u, v \in \mathbf{Z}$ , with  $\gcd(Z, w) = \gcd(u, v) = 1$ .

Expanding (12) produces three genus-two curves. One has no  $\mathbf{Q}_2$ -point because its primitive homogeneous sextic is always 5 modulo 8. A second has precisely the two rational points  $(-2, \pm 540)$ ; its Jacobian has rank one and the rational points are exhausted by Chabauty's method. Substitution gives  $r = -1/4$  and

$$\Phi(-1/4) = -\frac{491}{2^{14}},$$

which cannot be a reduced quotient of a fifth power by a seventh power.

### 3.2 The main genus-two curve

The remaining case gives the curve

$$C_1 : y^2 = x^6 - 6x^5 + 69x^4 - 38x^3 + 6x^2 - 240x + 1057. \quad (13)$$

Its Jacobian is torsion-free of rank three, and three explicit Mumford divisors form a  $\mathbf{Z}$ -basis. A Mordell–Weil sieve in the sense of [13], modulo  $1080J_1(\mathbf{Q})$ , places every rational point in one of the six classes represented by

$$\infty_+, \infty_-, (1/2, \pm 245/8), (3, \pm 65).$$

Exact formal-group calculations on these classes imply, for every affine  $(x, y) \in C_1(\mathbf{Q})$ ,

$$\text{ord}_3(x) \leq -3 \quad \text{or} \quad \text{ord}_5(y) \geq 1. \quad (14)$$

The point on (13) arising from (12) has

$$x = \frac{u}{v}, \quad y = \frac{(Z + w)^2 + 3w^2}{v^3}.$$

The case  $v = 0$  is also visible on the projective model. Coprimality forces  $u = \pm 1$ , while the two descent equations give

$$Z^2 + wZ - 3w^2 = \pm 1, \quad -w(Z + w) = 0.$$

If  $w = 0$ , then  $X = 0$ ; if  $Z = -w$ , then  $-3w^2 = \pm 1$ . Both are impossible under the hypotheses.

For  $v \neq 0$ , the second arm of (14) would imply  $5 \mid (Z + w)^2 + 3w^2$ , impossible for a primitive pair modulo 5. Hence  $27 \mid v$ . The second descent equation gives  $27 \mid w(Z + w)$ . Coprimality and (10) exclude  $27 \mid w$ , so  $27 \mid Z + w$ . Reducing the first descent equation modulo 3 now gives  $3 \mid u$ , contradicting  $\gcd(u, v) = 1$ .

**Proposition 3.1.** *The polynomial  $f_\eta$  has no rational root. Hence the sector  $[1, 6]$  is empty.*

## 4 The quadratic-factor sector

**Provenance.** The reduction to the fibre product and its six rational points is Dahmen–Siksek, Proposition 5.1 and Lemma 5.2 [1, Section 5]. We include the construction because it is the complete bridge from a quadratic root to the final fifth-power test. The database-free recovery of  $\mathbf{Q}(\sqrt{-35})$  from its local labels is the additional refinement used here.

Suppose that  $A_\eta$  has a quadratic direct factor  $L$ . Its real signature, ramification, and dyadic labels identify it without a field database.

Since  $f_\eta$  has exactly one real root,  $L$  is imaginary. At every prime outside  $\{3, 5, 7\}$ , the scaled local models are separable and every direct factor is unramified. At 2, the possible local degree patterns are  $[1, 2, 4]$  and  $[1, 3, 3]$ . A quadratic global direct factor must therefore be the unique unramified quadratic component in the first pattern. Thus 2 is inert in  $L$ . The negative odd fundamental discriminants supported on  $\{3, 5, 7\}$  and inert at 2 are  $-3$  and  $-35$ . Finally, the degree-two Newton-polygon component at 7 is ramified. Therefore

$$L = k = \mathbf{Q}(\omega). \quad (15)$$

Let  $u$  be a nonrational root of  $f_\eta$  in  $L$  and  $v = \bar{u}$ . Symmetric elimination gives a rational point on

$$D : \begin{cases} \beta^2 = \alpha^4 - 4\alpha^3 + 6\alpha^2 + 9, \\ \alpha(\alpha - 4) = -35\gamma^2. \end{cases} \quad (16)$$

Here

$$\alpha = \frac{(u + v)^2}{uv}.$$

**Proposition 4.1.** *The rational points on  $D$  are exactly*

$$(0, \pm 3, 0), \quad \left( \frac{35}{9}, \pm \frac{782}{81}, \pm \frac{1}{9} \right),$$

*with independent signs in the second family.*

*Proof.* Map  $D$  to the genus-two curve

$$-35y^2 = x(x-4)(x^4 - 4x^3 + 6x^2 + 9). \quad (17)$$

The étale algebra of the right side is  $\mathbf{Q} \times \mathbf{Q} \times L_4$ , where

$$L_4 = \mathbf{Q}(\vartheta), \quad \vartheta^4 - 4\vartheta^3 + 6\vartheta^2 + 9 = 0.$$

An exact 2-cover descent [12] gives three fake-Selmer classes, whose components at 0, 4, and  $\vartheta$  are

|         | 0   | 4 | $\vartheta$        |
|---------|-----|---|--------------------|
| $\xi_1$ | -3  | 1 | $3(\vartheta - 4)$ |
| $\xi_2$ | -35 | 1 | 1                  |
| $\xi_3$ | -35 | 1 | -5.                |

The first class cannot lift to a nonendpoint rational point on  $D$ . Indeed, in this class  $x = -3ta^2$  and  $x - 4 = tb^2$ , whereas the additional condition on  $D$  requires  $x(x-4)/(-35)$  to be a square. This would make  $3/35$  a square, which is impossible already over  $\mathbf{Q}_7$ . The endpoint  $x = 4$  does not lift either, since it would require  $\beta^2 = 105$ . The other two classes map to elliptic curves over  $L_4$ .

Their ranks are respectively 3 and 2, their torsion is trivial, and elliptic Chabauty [11] retains only the rational  $x$ -coordinates

$$x = 0, \quad x = 35/9.$$

For the rank-three curve, completeness is certified by saturation at every prime dividing the Chabauty index

$$6592880678883323322765152793600 = 2^{10} 3^5 5^2 7^4 \cdot 11 \cdot 17 \cdot 19 \cdot 29 \cdot 31 \cdot 37 \\ \cdot 43 \cdot 47 \cdot 53 \cdot 61 \cdot 71 \cdot 83 \cdot 97.$$

Substitution into (16) gives precisely the six points stated. □

If  $\alpha = 0$ , then  $v/u = -1$ . The root relation gives either  $u = 0$  or  $u = \pm\delta/5$ . The nonzero alternatives satisfy

$$\Phi(u) = \frac{7^4}{5^2},$$

which is not a reduced quotient of a fifth power by a seventh power. If  $\alpha = 35/9$ , then

$$\frac{v}{u} = \frac{17 \pm \delta}{18}.$$

The root relation gives either a branch value or the rational number

$$\frac{495535757310206240323260222}{432344158742282808792052775},$$

whose coprime numerator and denominator factor as

$$495535757310206240323260222 = 2^3 16^4 17 19^{10} 23, \\ 432344158742282808792052775 = 5^2 29^7 139^7.$$

The numerator is not a fifth power. This proves:

**Proposition 4.2.** *The sector  $[2, 5]$  is empty.*

## 5 The cubic–quartic sector

**Provenance.** The elimination follows Dahmen–Siksek, Section 7, in particular Proposition 7.1 and Lemmas 7.2–7.7 [1, Section 7]. Our additional contribution is to certify the ramified local step inside the intrinsic integral formal neighbourhood, rather than treating model-dependent coordinates as an unexamined proxy for the formal group.

The quartic direct factor is totally imaginary and unramified outside  $\{3, 5, 7\}$ . Dahmen–Siksek consider the complete lists of nineteen cubic fields of signature  $(1, 1)$  and seventeen quartic fields of signature  $(0, 2)$  unramified outside this set. They compare each cubic–quartic product with the permitted local algebras at 3, 5, and 7. Their Proposition 4.3 leaves two products, whose cubic factors are  $\mathbf{Q}(\sqrt[3]{5^\alpha 7})$  for  $\alpha = 1, 2$ , and whose quartic factor is the same field in both cases [1, Proposition 4.3].

The surviving field is

$$M = \mathbf{Q}[a]/(a^4 - a^3 + 3a^2 + 3a + 9) = k\left(\sqrt{\frac{7+\delta}{2}}\right). \quad (18)$$

The associated quotient is the genus-two curve

$$C_2 : \quad V^2 = U(U-4)(U^4 - 4U^3 + 6U^2 + 9) \quad (19)$$

over  $k$ . Write  $J_2 = \text{Jac}(C_2)$ .

Here is the solution-bearing condition used in the terminal argument. Put  $\kappa = (7 + \delta)/2$ , choose a root  $u$  generating  $M = k(\sqrt{\kappa})$ , and let  $v$  be its conjugate over  $k$ . The quotient map of [1, Lemma 7.2] gives a point  $P \in C_2(k)$  with

$$U(P) = \frac{(u+v)^2}{uv}, \quad \rho(P) = \Phi(u) = \eta \in \mathbf{Q},$$

where  $\rho \in k(C_2)$  is the quotient parameter. The denominator  $uv$  is nonzero because  $\eta \neq 0$ . Moreover  $U(P) \notin \mathbf{Q}$ . Indeed,  $N_{k/\mathbf{Q}}(\kappa) = 21$  is not a square in  $k$ , so  $M/\mathbf{Q}$  is not Galois and its only quadratic subfield is  $k$ . If  $U(P) \in \mathbf{Q}$ , the equation  $(u/v)^2 + (2 - U(P))(u/v) + 1 = 0$  implies  $u/v \in k$ . Since  $uv \in k$ , we obtain  $u^2 \in k$ , and hence  $u = b\sqrt{\kappa}$  for some nonzero  $b \in k$ . The coefficient of  $\sqrt{\kappa}$  in  $\Phi(u) \in k$  then gives  $b^2 = -7/(5\kappa)$ . This is impossible: its norm would give  $N_{k/\mathbf{Q}}(b)^2 = 7/75$ , which is not a rational square.

### 5.1 The Mordell–Weil group and sieve

We use the Mordell–Weil group determination of [1, Lemma 7.3], which gives

$$J_2(k) = \langle D_0 \rangle_{10} \oplus \mathbf{Z}D_1 \oplus \mathbf{Z}D_2 \oplus \mathbf{Z}D_3. \quad (20)$$

For reproducibility, we also give a direct finite-field certificate for the remaining 2-saturation in this statement. Put  $A = J_2(k)$ ,  $H = \langle D_0, D_1, D_2, D_3 \rangle$ , and  $H_0 = \langle D_0, D_1, D_2, D'_3 \rangle$ . The rational and twisted Mordell–Weil computations used in the cited lemma give  $\text{rank } A = 3$ ; good reduction at 11 and 13, together with the displayed order of  $D_0$ , gives  $A_{\text{tors}} = \langle D_0 \rangle \simeq \mathbf{Z}/10\mathbf{Z}$ . For the nontrivial  $\sigma \in \text{Gal}(k/\mathbf{Q})$  and every  $P \in A$ ,

$$2P = (P + \sigma P) + (P - \sigma P).$$

The full rational and twisted groups identify the fixed and anti-fixed summands on the right with the named generators of  $H_0$ . Thus  $2A \subseteq H_0 \subseteq H$ , the second inclusion following from the verified relation

$$D'_3 = 9D_0 + 3D_1 - D_2 - 2D_3.$$

It therefore remains only to show that the four displayed classes span  $A/2A$ .

At the good prime  $\mathfrak{l} = (173, \delta - 22)$ , the change of variables

$$u = -36/U, \quad v = 1296V/U^3$$

puts the curve into the odd monic form

$$v^2 = u^5 + 9u^4 + 864u^3 + 28512u^2 + 373248u + 1679616.$$

Its five roots modulo  $\mathfrak{l}$ , in the certified order, are 15, 90, 119, 122, 164. Applying the usual odd-degree  $u - T$  Kummer map and then the five quadratic characters of  $\mathbf{F}_{173}^\times$  to  $D_0, D_1, D_2, D_3$  gives the matrix

$$\begin{pmatrix} 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}.$$

The determinant of its first four rows is 1 over  $\mathbf{F}_2$ . Hence these four classes are independent in  $A/2A$ . Since  $\dim_{\mathbf{F}_2} A/2A = 3 + 1 = 4$ , they form a basis; as  $A/H$  was already killed by 2, this proves  $A = H$ . The archive contains both the short reconstruction and an independent evaluation of every Kummer entry.

A Mordell–Weil sieve [13] with modulus 2240 yields twelve points  $P_1, \dots, P_{12}$  such that every  $Q \in C_2(k)$  satisfies

$$[Q - P_i] \in 2240J_2(k) \tag{21}$$

for a unique  $i$ .

Let  $\mathfrak{P}$  be the unique prime of  $k$  above 7, let  $k_{\mathfrak{P}}$  be the completion, and put  $R = \mathcal{O}_{k_{\mathfrak{P}}}$ ,  $\mathfrak{m} = (\delta)$ . Thus

$$v_{\mathfrak{P}}(\delta) = 1, \quad v_{\mathfrak{P}}(7) = 2.$$

Use Flynn’s integral  $\mathbf{P}^{15}$  model of  $J_2$  with origin  $(1 : 0 : \dots : 0)$  and normalized coordinates  $s_i = a_i/a_0$ . On the origin neighbourhood  $\mathcal{N}$ , the projective-coordinate framework of [8] and Corollary 2.2 and Theorem 3.5 of [7] identify  $(s_1, s_2)$  as integral formal parameters and give an integral formal group law. Define

$$G[2] = \{A \in \mathcal{N} : s_1(A), s_2(A) \in \mathfrak{m}^2\}. \tag{22}$$

It is a subgroup, and the coordinate identities place every nonorigin normalized coordinate in  $\mathfrak{m}^2$ .

The exact origin-chart valuations of  $2240D_1$ ,  $2240D_2$ , and  $2240D_3$  in the adapted integral coordinates described in Appendix B are

$$\begin{array}{l|l} D_1 & 0, 2, 2, 4, 4, 4, 6, 6, 6, 6, 8, 8, 8, 8, 8, 16 \\ D_2 & 0, 5, 3, 10, 8, 6, 15, 13, 11, 9, 20, 18, 16, 14, 12, 20 \\ D_3 & 0, 2, 2, 4, 4, 4, 6, 6, 6, 6, 8, 8, 8, 8, 8, 16. \end{array} \tag{23}$$

The torsion generator is killed by 2240, so

$$2240J_2(k) \subset G[2]. \tag{24}$$

## 5.2 The twelve local classes

For each  $P_i$ , exact pullback calculations prove

$$[Q - P_i] \in G[2] \implies Q \equiv P_i \pmod{\mathfrak{P}^2}. \tag{25}$$

At each of five regular finite bases, the 98 depth-two congruence classes split into one base class and 97 rejected classes. At each of the five ramified finite bases, the complete tree is

| $n$ | tested | base | rejected | unresolved | 49-digit lifts | admissible at $n + 1$ |
|-----|--------|------|----------|------------|----------------|-----------------------|
| 2   | 98     | 1    | 91       | 6          | 294            | 294                   |
| 3   | 294    | 0    | 252      | 42         | 2058           | 686                   |
| 4   | 686    | 0    | 588      | 98         | 4802           | 686                   |
| 5   | 686    | 0    | 686      | 0          | --             | --                    |

The last two columns make the branching arithmetic explicit. Each unresolved pair modulo  $\mathfrak{P}^n$  has  $7^2 = 49$  two-coordinate digit lifts; the next row counts only those lifts that also satisfy the curve equation modulo  $\mathfrak{P}^{n+1}$ . At the ramified special-fibre point this number is not uniformly seven, which is why the tested-node column is not obtained by multiplying the unresolved column by a fixed branching factor. The singular special-fibre point  $U \equiv 4 \pmod{\mathfrak{P}}$  is included in this calculation. At the two points at infinity, the opposite branch lies outside  $\mathcal{N}$  and the identity branch has

$$s_1 = \pm \frac{1}{2}w + O(w^2),$$

with unit linear coefficient. Here a node is a full congruence class, and a rejection is a uniform valuation statement on that class; no finite sample is used as an exhaustion argument.

Combining (21), (24), and (25), every  $k$ -point lies in one of the twelve explicit  $\mathfrak{P}^2$ -disks. Exact 7-adic and 23-adic conditions, followed where necessary by Coleman integration, show that no disk contains a solution-bearing point beyond its reference  $P_i$ . The reference points themselves either have parameter 0, 1, or  $\infty$ , or fail the required rational admissibility condition; hence none yields a nonzero primitive solution.

**Proposition 5.1.** *The sector  $[3, 4]$  is empty.*

## 6 The irreducible septic sector

**Provenance.** The seven-field enumeration is Putz's classification theorem. The elimination of the six pure fields and the exceptional field is the new terminal part of the argument.

Suppose now that  $A_\eta$  is a field. It has degree 7, signature  $(1, 3)$ , is unramified outside  $\{3, 5, 7\}$ , and has the prescribed local algebras at 2, 3, 5, and 7 of [1, Corollary 3.4 and Propositions 3.5–3.7]. Putz's exhaustive Hunter enumeration [2, Theorem 3.50] gives exactly seven possibilities:

$$\mathbf{Q}(\sqrt[7]{3^a 5}) \quad (1 \leq a \leq 6), \quad \mathbf{Q}[T]/(T^7 - 483T^2 + 3955T - 3945). \quad (26)$$

We call the first six fields pure and the last field exceptional.

### 6.1 The pure fields

The Fano-resolvent construction and the rational-parameter theorem in this subsection supply the geometric obstruction that was absent from the cited prior work.

The normal closure of a pure field has Frobenius group  $F_{42}$  and unique quadratic subfield

$$K_7 = \mathbf{Q}(s), \quad s^2 = -7.$$

#### 6.1.1 The Fano-resolvent bridge

The passage from a pure septic field to a curve is as follows. Scale  $V = 15T$  and write

$$h_t(V) = V^7 - 35V^6 + 315V^5 - 15^6 t, \quad t = \eta = \frac{X^5}{Z^7}. \quad (27)$$

Let  $v_1, \dots, v_7$  be its roots. A labelled Fano plane  $F$  on these seven roots has seven lines, each a three-element subset; define

$$M_F = \sum_{\{i,j,k\} \text{ a line of } F} v_i v_j v_k, \quad q_F = \frac{M_F}{225}. \quad (28)$$

There are thirty labelled Fano planes. They form two  $A_7$ -orbits  $\mathcal{F}_+$  and  $\mathcal{F}_-$  of fifteen planes. Since

$$\text{disc}(h_t) = -7^7 15^{36} t^4 (t-1)^2,$$

the two orbit polynomials are conjugate over  $K_7$ . After fixing the Vandermonde orientation, put

$$R_+(q, t) = \prod_{F \in \mathcal{F}_+} (q - q_F) = \frac{U(q, t) + sW(q, t)}{2}, \quad U, W \in \mathbf{Z}[q, t]. \quad (29)$$

The normalization of  $R_+ = 0$  is denoted  $C_+$ . On its fifteen sheets the branch permutations above  $t = 0, 1, \infty$  have cycle structures

$$5^3, \quad 3^5, \quad 7^2 1.$$

Riemann–Hurwitz therefore gives  $g(C_+) = 3$ .

To see why this quotient detects the six pure fields, call  $F \in \mathcal{F}_+$  and  $G \in \mathcal{F}_-$  compatible when they have no line in common. We use unordered pairs  $\{F, G\}$ , since odd permutations exchange the two  $A_7$ -orbits. Each  $F$  has exactly eight compatible  $G$ 's, giving 120 such pairs. The normalization of the corresponding cover is a curve  $X_{42}/\mathbf{Q}$ . The stabilizer of an unordered compatible pair in  $A_7$  is the Frobenius group  $F_{21}$ , and its stabilizer in  $S_7$  is  $F_{42}$ . Thus  $X_{42}$  has degree 120 over the  $t$ -line and, after base change to  $K_7$ , the forgetful map

$$X_{42, K_7} \longrightarrow C_+ \quad (30)$$

has degree 8. The branch permutations on the 120 sheets have cycle structures  $5^{24}$ ,  $3^{40}$ , and  $7^{17} 1$ , so  $g(X_{42}) = 20$ .

If (27) cuts out one of the pure fields  $\mathbf{Q}(\sqrt[7]{3^a 5})$ , its normal closure has group  $F_{42}$ . The fixed-field/resolvent correspondence supplies a rational point on  $X_{42}$ , hence a  $K_7$ -point on  $C_+$  through (30). Both maps preserve  $t$ . A nonzero solution has  $t \notin \{0, 1, \infty\}$ , so the resulting point is not in a branch fibre. This constructs the claimed bridge rather than merely inferring it from matching Galois groups.

The canonical model of  $C_+$  is the explicit smooth plane quartic printed in Appendix A. Its smoothness proves again that its genus is 3. The inverse parameter is the exact rational function  $t = -G_t/H_t$ ; all coefficients of the two quintics are printed in Table 2. The archive file `p7_f42_canonical_exact.json.gz` specifies the forward coordinates  $N_j(q, t)$  for  $j = 1, 2, 3$ , and the inverse forms  $G_q, H_q$ . Exact reduction modulo  $R_+(q, t)$  verifies

$$Q(N_1, N_2, N_3) = 0, \quad G_q(N) + qH_q(N) = 0, \quad G_t(N) + tH_t(N) = 0,$$

where  $Q$  is the printed quartic and the two inverse denominators are nonzero in the function field. Recovering both generators  $q$  and  $t$  proves that this map induces an isomorphism of function fields, hence of the smooth projective models, preserving the parameter.

**Theorem 6.1** (rational-parameter theorem). *Let  $J_+ = \text{Jac}(C_+)$ . Then*

$$\{Q \in C_+(K_7) : t(Q) \in \mathbf{P}^1(\mathbf{Q})\} = \{P_1, P_2, P_3, P_4, P_5\}.$$

*The corresponding parameter values are  $0, 1, \infty, \infty, \infty$ .*

*Proof.* Put  $D_i = [P_i - P_1]$  and  $H_0 = \langle D_2, D_3, D_4, D_5 \rangle$ . Appendix D gives the complete containing-space and local stopping argument in the genus-three framework of [9], including the 7-adic kernel witness and the literal 5-adic obstruction. Together with the independently checked lower bound, it proves

$$\dim_{\mathbf{F}_2} \text{Sel}_2(J_+/K_7) = 4, \quad \text{rank } J_+(K_7) = 4, \quad \text{Sha}(J_+/K_7)[2] = 0.$$

The independent 2-saturation witness shows that  $D_2, D_3, D_4, D_5$  form a basis modulo 2. Thus  $H_0$  has odd index. The existence of the fifth-division class used below can be seen from the branch divisor. Write  $A = (t)_0/5$ , a  $K_7$ -rational divisor of degree three, and set  $B = [A - 3P_1]$ . The pole divisor is  $(t)_\infty = 7P_3 + 7P_4 + P_5$ , so

$$5B = 7D_3 + 7D_4 + D_5.$$

Consequently the class  $E = 3B - 4D_3 - 4D_4$  satisfies

$$5E = D_3 + D_4 + 3D_5. \quad (31)$$

Set  $H_1 = \langle D_2, E, D_4, D_5 \rangle$ . Equation (31) gives  $H_0 \subseteq H_1$ . Four independent local  $C_{25}$  rows have determinant 4 modulo 5, and  $J_+(K_7)[5] = 0$ ; hence  $H_1$  is 5-saturated. Its index is therefore coprime to 400, and  $H_1 \rightarrow J_+(K_7)/400J_+(K_7)$  is surjective. This is the coefficient coverage used in the sieve.

At the split primes

$$23, 43, 67, 71, 79, 107, 109, 113,$$

both reductions of the curve are enumerated, as are the images of every reduced point in the relevant local quotients of  $J_+$ . The same global coefficient vector must occur at both places. In addition, rationality of  $t(Q)$  imposes equality of the two projective reductions of  $t$ . The resulting finite sieve has the following exact counts:

| stage                               | surviving coefficient classes |
|-------------------------------------|-------------------------------|
| coupled 2-primary sieve modulo 16   | 592                           |
| 5-primary prefilter modulo 25       | 1252                          |
| joint point fingerprints modulo 400 | 144                           |

The final classes use exactly the five vectors modulo 25

$$(0, 0, 0, 0), (1, 0, 0, 0), (0, 5, 24, 22), (0, 0, 1, 0), (0, 0, 0, 1),$$

and their projection at the two primes above 23 consists precisely of the five ordered residue pairs of  $P_1, \dots, P_5$ .

The combined 23-adic logarithm matrix has size  $6 \times 4$  and rank 4. The resulting two independent logarithmic functionals vanish on  $H_0$ . Since  $H_0$  has finite index, they vanish on all of  $J_+(K_7)$ : for any  $P \in J_+(K_7)$ , a nonzero integer multiple of  $P$  lies in  $H_0$ , and the logarithm takes values in characteristic zero. At the disks of  $P_2, P_3, P_4, P_5$ , the Siksek tangent determinants modulo 23 are

$$12, \quad 16, \quad 4, \quad 19. \quad (32)$$

This is the number-field Chabauty criterion of [10]; the values are units, so each disk contains only its reference point among common Coleman zeros.

At  $P_1$ , write the two local parameters as  $23u, 23v$ , so that the whole product residue disk is parametrized by  $(u, v) \in \mathbf{Z}_{23}^2$ . For the two annihilator integrals  $F_1, F_2$ , use the normalized equations  $F_1/23 = 0$  and  $(F_2 - 6F_1)/23^2 = 0$ . The second numerator is integrally divisible by

$23^2$ : its linear coefficients vanish modulo 23 before scaling, and all higher terms supply the extra factor. The reductions are

$$2u + 4v = 0, \quad 21u + 6v + 14u^2 + 21v^2 = 0.$$

They have exactly two simple reduced zeros,

$$(u, v) = (0, 0), \quad (14, 16),$$

with determinants 20 and 3. Multivariate Hensel lifting gives one exact common zero in each of these two classes and none elsewhere. The first is  $P_1$ . The arithmetic retains seven digits after both normalizing divisions: an omitted integrated term of exponent  $k \geq 12$  has valuation at least  $k - v_{23}(k) \geq 12$ , and the coefficient arithmetic retains the required precision. Multiplication by 23 gives the local coordinates modulo  $23^8$ . The canonical parameter denominator is a unit at both lifted endpoints, so its evaluation preserves those eight digits. At the second zero, the two parameter values modulo  $23^8 = 78,310,985,281$  are

$$t_{s=4} = 59,233,664,629, \quad t_{s=19} = 24,239,267,738.$$

Their difference has exact 23-adic valuation 5, with first nonzero quotient digit 9. Hence this common Coleman zero does not have rational  $t$ . The five reference points are therefore the entire rational-parameter locus.  $\square$

Each  $P_i$  in Theorem 6.1 lies above one of the branch values 0, 1, and  $\infty$ . A nonzero primitive solution would give a point with rational parameter outside this set. Consequently none of the six pure fields occurs.

## 6.2 The exceptional field

The general modularity and conductor theorems used here are cited inputs. The project-specific specialization, exhaustive packet and ray-character interfaces, and the terminal  $q = 29$  comparison form the new finite elimination.

We use the Frey abelian variety of Pacetti and Villagra Torcomian [3] for

$$a^5 + b^p + c^3 = 0. \tag{33}$$

In our variables

$$(a, b, c) = (X, -Z, Y), \quad t_0 = -\frac{a^5}{c^3} = \frac{\eta}{\eta - 1}. \tag{34}$$

Put  $F = \mathbf{Q}(\sqrt{5})$  and let  $\bar{\rho}_7 : G_F \rightarrow \mathrm{GL}_2(\bar{\mathbf{F}}_7)$  be the residual representation of the weight-two realization of their plus motive at  $t_0$ . The use of the modular method at the small exponent  $p = 7$  rests on the following precise inputs, not on the headline large-exponent theorem of [3]. Theorem 2.4(1) of [3] states that for  $r = 3$  and  $q \geq 5$  the plus motive is modular for every rational specialization. Its determinant is the mod-7 cyclotomic character (equivalently, the compatible system has trivial Nebentypus), and Theorem 2.1(3)–(4) gives the needed ramification control and finiteness at every prime above 7. Here 7 is inert and unramified in  $F$ , so there is a single such prime.

Now assume that  $\bar{\rho}_7$  is irreducible. Apply [3, Theorem 7.8] with  $p = 7$  to the specialization (34). This specialized theorem incorporates modularity, the local conductor calculation, and Hilbert level lowering; its proof cites [4] for the latter. It gives a parallel-weight-two Hilbert newform over  $F$ , with trivial Nebentypus, whose residual representation is  $\bar{\rho}_7$  and whose level is one of

$$3^{\epsilon_3}(\sqrt{5})^{\epsilon_5}, \quad (\epsilon_3, \epsilon_5) \in \{2, 3\}^2. \tag{35}$$

The conclusion here is a newform conclusion, but the finite elimination below does not depend on reconstructing a new quotient. Let  $V_N$  be the full mod-7 cuspidal Brandt module at one of the four levels  $N$  in (35). Our earlier computation first imposed eighteen local trace-polynomial conditions, so its output must not be called the full ambient module. We instead verify the coverage of every condition.

Section 7.4 of [3] gives an exhaustive trichotomy at an auxiliary prime: ordinary specialization, one of the two degenerations  $t = 0, \infty$ , or the level-lowering specialization  $t = 1$ . From the complete upstream `Data.txt`, the certificate independently reduces all 6253 trace polynomials at 37 primes modulo 7. Separately, it reconstructs all 1264 ordinary parameters at those 18 filter primes and all 128 boundary character possibilities at the 16 split primes. It performs the required relative-degree-two trace transport for the boundary data and adjoins both signs of  $N(\mathfrak{q}) + 1$ . Seventeen of those eighteen filters are exactly the resulting squarefree allowed-trace polynomials. At  $\ell = 131$  the degree-48 filter in our earlier computation omitted the allowed level-lowering trace 1, since  $-(131 + 1) \equiv 1 \pmod{7}$ . We therefore replace only that condition, in memory, by  $T^{49} - T$ , which admits every  $\mathbf{F}_{49}$ -valued trace.

Write  $W_N \subseteq V_N$  for the intersection after these seventeen verified conditions and the condition at 131 imposing no further restriction on  $\mathbf{F}_{49}$ -valued Frey eigenvalues. Characteristic-zero Jacquet–Langlands [6, Theorem 3.9 and §4] supplies the corresponding Hecke eigenvector in the definite quaternionic function model. The exact order and neighbour computations verify the maximal order, class number one, and the complete Hecke operators; all finite stabilizer orders divide 120, hence are prime to 7. The integral function lattice and its cuspidal mass condition are therefore preserved at primes above 7. Scale the eigenvector to be primitive in this local lattice and reduce. The resulting nonzero vector has the required residual eigenvalues. The local trichotomy places it in  $W_N$ . This proves coverage without degeneracy-map subtraction. A fresh simultaneous primary decomposition has packet dimensions, with multiplicity, summing to  $\dim W_N$  at every level. The exceptional field is unramified at 2 with local factor degrees  $[1, 2, 4]$ . If  $2 \mid Z$ , the integral equation for  $W = ZT$  reduces to

$$W^7 + 1 = (W + 1)(W^3 + W + 1)(W^3 + W^2 + 1) \quad \text{over } \mathbf{F}_2.$$

Its factors are distinct and irreducible, giving local degrees  $[1, 3, 3]$ , a contradiction. Hence  $Z$  is odd, and exactly one of  $X, Y$  is even. In the variable convention  $X^5 + Y^3 = Z^7$  used here, the prime-2 calculation gives the necessary residual trace  $-1$  when  $Y$  is even and  $0$  when  $X$  is even. Retaining every packet for which that scalar is a root of the residual  $T_2$  polynomial gives

|             | 22 | 23 | 32 | 33  |
|-------------|----|----|----|-----|
| $\dim W_N$  | 8  | 38 | 28 | 98  |
| all packets | 4  | 14 | 12 | 24  |
| $Y$ -even   | 1  | 2  | 2  | 5   |
| $X$ -even   | 2  | 10 | 3  | 11. |

Across all thirty-six retained packet occurrences, the set of residual  $T_{\mathfrak{q}}$  annihilators at  $\mathfrak{q} \mid 29$  is exactly

$$T, \quad T^2, \quad T + 2, \quad T^2 + 5T + 2, \quad T^2 + 3T + 4.$$

The terminal list already contains all five, because  $T + 2 = T - 5$  modulo 7; together with  $T - 2$  it is the same six-factor list used in the resultant calculation below. Thus every residual representation supplied by level lowering survives the preceding filters and is covered before the terminal comparison. The old/new calculation remains only a diagnostic. We make no complete-newspace claim and require no mod-7 injectivity or semisimplicity statement for degeneracy maps.

Suppose instead that  $\bar{\rho}_7$  is reducible. Its semisimplification has the determinant-compatible form

$$\bar{\rho}_7^{\text{ss}} = \psi \oplus \psi^{-1} \bar{\chi}_7. \tag{36}$$

At a finite prime  $\lambda \nmid 7$ , the cyclotomic character is unramified, so the two characters in (36) have the same Artin-conductor exponent. The conductor of the semisimplification is at most that of the representation. Thus the local conductor bounds at the primes above 3 and 5 give

$$2a_\lambda(\psi) = a_\lambda(\bar{\rho}_7^{\text{ss}}) \leq a_\lambda(\bar{\rho}_7) \leq 3, \quad a_\lambda(\psi) \leq 1.$$

At every other finite prime away from 7, the character  $\psi$  is unramified.

The prime  $\mathfrak{p} = (7)$  is unramified and inert in  $F$ , with residue field  $\mathbf{F}_{49}$ . By the finite-flat character classification [5, Corollary 3.4.4], its restriction to inertia has the form

$$\psi|_{I_{\mathfrak{p}}} = \omega_2^{r_0 + 7r_1}, \quad r_0, r_1 \in \{0, 1\},$$

The character has level dividing two because it is a rank-one character of the full local Galois group. Here  $\omega_2$  is a fundamental character of level two, normalized so that  $\omega_2^8 = \bar{\chi}_7|_{I_{\mathfrak{p}}}$ . The two mixed possibilities must be excluded globally. Put  $\varepsilon = (1 + \sqrt{5})/2$  and

$$u = \varepsilon^8 = 13 + 21\varepsilon.$$

The unit  $u$  is totally positive and satisfies

$$u \equiv 1 \pmod{3(\sqrt{5})}, \quad u \equiv -1 \pmod{7}.$$

Global reciprocity applied to  $u$  shows that the value of  $\psi$  on its local reciprocity image at  $\mathfrak{p}$  is 1: at every other finite place this follows from the conductor bounds just proved, and at both real places it follows from total positivity. The displayed inertial character gives that value as  $(-1)^{r_0 + r_1}$ . Hence  $r_0 = r_1$ . If both are zero then  $\psi$  is unramified at  $\mathfrak{p}$ ; if both are one then  $\psi^{-1}\bar{\chi}_7$  is unramified there. Interchanging the two characters if necessary, we may therefore choose  $\psi$  such that

$$\text{cond}(\psi) \mid 3(\sqrt{5}). \quad (37)$$

Including both real places gives a modulus containing the conductor of  $\psi$  regardless of its signs. Its narrow ray group is  $C_4 \times C_2$ , so there are eight possible characters, all of which are included in the comparison.

At  $q = 29$ , take  $\mathfrak{q} = (29, \sqrt{5} - 11)$ ; the conjugate prime gives the conjugate packet and the same common-root test. The exceptional septic polynomial is squarefree modulo 29 with factor degrees  $[1, 3, 3]$ . Among  $\eta \in \mathbf{F}_{29} \setminus \{0, 1\}$ , the polynomial  $\Phi(T) - \eta$  has this pattern precisely for  $\eta = 10, 14, 24, 28$ .

The branch residues do not give this pattern. If  $29 \mid X$ , scale the five-root cluster at  $T = 0$  using  $\text{ord}_{29}(\eta) = 5 \text{ord}_{29}(X)$ . Its separable reduction has degrees  $[1, 2, 2]$ , since  $29 \equiv -1 \pmod{5}$ ; the complementary quadratic has degrees  $[1, 1]$ . If  $29 \mid Y$ , the corresponding three-root cluster at  $T = 1$  has degrees  $[1, 2]$ , and the complementary quartic has degrees  $[1, 1, 2]$ . Both cases therefore have pattern  $[1, 1, 1, 2, 2]$ . If  $29 \mid Z$ , the scaled coordinate  $U = ZT$  gives reduction  $15U^7 - X^5$ ; its pattern is  $[1, 1, 1, 1, 1, 1, 1]$  or  $[7]$  because  $\mathbf{F}_{29}$  contains the seventh roots of unity. Hensel's lemma applies to each scaled separable reduction. Thus only the four ordinary values remain. By (34) they correspond to  $t_0 = 14, 10, 25, 15$ . Direct point counting on

$$y^2 = 5x^6 - 12x^5 + 10t_0x^3 + t_0^2$$

gives the following polynomials over  $\mathbf{F}_7$  satisfied by the two conjugate real-multiplication Frobenius traces. Explicitly, if  $N_1 = \#C_{t_0}(\mathbf{F}_{29})$ ,  $N_2 = \#C_{t_0}(\mathbf{F}_{29^2})$ ,  $A = 30 - N_1$ , and  $B = (A^2 - (842 - N_2))/2$ , the degree-four Frobenius polynomial is  $X^4 - AX^3 + BX^2 - 29AX + 29^2$ , and the two traces satisfy  $T^2 - AT + (B - 58)$ . The table records the latter polynomial modulo 7:

| $\eta$ | $t_0$ | curve polynomial |
|--------|-------|------------------|
| 10     | 14    | $T^2 + 6T + 6$   |
| 14     | 10    | $T^2 + 6T + 4$   |
| 24     | 25    | $T^2 + 4$        |
| 28     | 15    | $T^2 + 2T + 3$ . |

(38)

For clarity, the next list is an *annihilating family*; its members are not asserted to be irreducible or squarefree. For each of the four ambient cuspidal spaces, reduce the characteristic polynomial of the Hecke operator  $T_q$  modulo 7 on every packet surviving the conservative  $T_2$  condition. The preceding dimension accounting proves that every residual eigenvalue  $a_q(f) \bmod \mathfrak{p}$ , for every relevant newform and each coefficient prime  $\mathfrak{p} \mid 7$  realizing the required Frey congruence, is a root of one of the first four or the sixth displayed factors. We retain the factors as they occur in the packet calculation, so they need not be squarefree or minimal. The reducible characters contribute the possible traces  $\psi(\mathfrak{q}) + \psi(\mathfrak{q})^{-1}$ . The resulting complete annihilating family is

$$T, \quad T^2, \quad T^2 + 5T + 2, \quad T^2 + 3T + 4, \quad T - 2, \quad T - 5. \quad (39)$$

Here  $T^2$  and  $T^2 + 3T + 4 = (T - 2)^2$  deliberately record multiplicity; for the resultant test they have the same roots as  $T$  and  $T - 2$ . For the specified prime  $\mathfrak{q}$ , the ray class has order two: a totally positive generator is  $6 - \varepsilon$ , whose square is congruent to the totally positive unit  $\varepsilon^2$  modulo  $3(\sqrt{5})$ ; its class is nontrivial, since modulo 3 the generator is  $-\varepsilon = \varepsilon^5$ , whereas every totally positive unit is an even power of  $\varepsilon$ , whose order modulo 3 is 8. Thus the reducible traces are 2 and  $-2$ , covered by the displayed linear factors. The 24 pairwise resultants of (38) and (39) form the matrix

$$\begin{pmatrix} 6 & 1 & 5 & 1 & 1 & 5 \\ 4 & 2 & 3 & 1 & 6 & 3 \\ 4 & 2 & 6 & 1 & 1 & 1 \\ 3 & 2 & 6 & 2 & 4 & 3 \end{pmatrix} \pmod{7}. \quad (40)$$

Every entry is nonzero. Thus the exceptional field is incompatible with both the irreducible and reducible residual alternatives.

**Proposition 6.2.** *The sector [7] is empty.*

## 7 Proof of the main theorem

*Proof of Theorem 1.1.* Suppose that a nonzero primitive solution exists and reorder it as in (2). The descent algebra  $A_\eta$  belongs to exactly one of the four sectors in (7). Proposition 3.1 excludes [1, 6]. Proposition 4.2 excludes [2, 5]. Proposition 5.1 excludes [3, 4]. Proposition 6.2 excludes [7]. Every possible factorization is therefore impossible, a contradiction.  $\square$

## 8 Computational certification

The computer-assisted portions of the proof are finite statements over  $\mathbf{Q}$ , number fields, finite fields, and truncated local rings. They are organized so that discovery is not used as evidence: every accepted output is checked against exact input data and against the mathematical predicate stated in the relevant proposition. The certificate archive serves two different purposes. It independently reconstructs finite calculations from the cited arguments of Dahmen–Siksek, and it supplies the load-bearing arithmetic for the new seven-field elimination. The provenance column below keeps those purposes distinct.

### 8.1 Exact arithmetic layer

The exact layer checks polynomial identities, resultants, discriminants, field embeddings, ideal-class representatives, divisor relations, point incidence, and rational substitutions. Integer and rational calculations are repeated without floating-point arithmetic. Manifests identify the input bytes and the corresponding outputs.

## 8.2 Finite exhaustion layer

Whenever a finite search is used as an exhaustion, the searched set is defined before enumeration. This applies to local factor patterns, Selmer classes, reduced curve points, Mordell–Weil coefficient classes, residue trees, ray characters, newform packets, and resultant comparisons. A residue-tree node denotes a complete congruence class. A rejected node is accompanied by a uniform congruence or valuation obstruction; it is not a sample point.

## 8.3 Local analytic layer

Formal-group and Coleman calculations record their working precision, unit pivots, ranks, determinants, and higher-precision replays. Divisions are made only by certified units or with their exact loss of precision recorded. Simple-root conclusions are accompanied by a nonzero reduced Jacobian determinant. Singular classes are subdivided or normalized until the stated conclusion follows.

## 8.4 Software and trust boundary

The calculations use PARI/GP [15] for exact number-field arithmetic, SageMath [16] for curve and finite-field calculations, and Magma [14] for descent, Mordell–Weil, saturation, newform, and Chabauty routines. Their role is the same as that of large integer arithmetic or a table of modular forms in a conventional proof: the inputs, outputs, and verification conditions are part of the mathematical record. The argument also imports the general theorems cited in the bibliography, rather than attempting to reprove them from first principles. In the cubic–quartic sector, the rational and twisted Mordell–Weil and torsion results in [1, Lemma 7.3] remain explicit inputs. The finite-field 173 Kummer argument supplies the separate 2-saturation step; it does not replace those rank and torsion inputs. Seven archived Magma inputs, including both full-group computations, were executed through the official Magma calculator, version V2.29-10. Their exact inputs, outputs and mathematical checks are included in the companion verification materials. In both full-group computations, the returned `finite_index` and `proved` flags are true, and the returned rank bounds are respectively 1 and 2; the scripts also verify the named generators in the returned groups. These executions use Magma’s algorithms; they are not independent implementations. The numerical height comparison supplements, and does not replace, the separate exact height bounds.

**Substantial AI-assisted research and writing.** OpenAI Codex was used extensively in developing and checking mathematical arguments, computational investigation, program and certificate development, verification design, debugging, release preparation, and drafting and revising the manuscript. OpenAI ChatGPT also supported the research and writing. Anthropic Claude acted as an independent auditor, reproducing the finite data from published inputs and reviewing every draft; its findings led to the database-free identification of  $\mathbf{Q}(\sqrt{-35})$  and to the model-intrinsic formal-group argument at the prime above 7. Peter Chocian originated the project’s discovery approach and research direction, supplied key inputs, reviewed the evidence, and accepts responsibility for all claims. AI-generated suggestions and reviews are not mathematical certificates: the proof rests on the arguments given here, the cited results and the specified computations. These AI-assisted reviews do not constitute independent human peer review or proof-assistant formalization.

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## 8.5 Companion certificate archive

The programs, exact inputs, certificates and verification records are available from the repository [bbpcho/primitive-357-proof-audit](#). The replacement computational companion is `PRIMITIVE_357_REPLAY_COMPANION_2026-09-15_V2.zip` (320731221 bytes). Its public release tag is

`replay-companion-2026-09-15.2.`

Its SHA-256 digest is

`a7e8308e906ab21410a18748e7b253518b26892aeab374baae7044407b164d85`

This replaces the two withdrawn archives with historical tags

`verified-2026-09-14.1`  
`first-submission-2026-09-15.1.`

The current paper and its small arXiv source archive are separate assets; earlier manuscript copies inside the computational records are retained as provenance.

The companion stores repeated evidence once and reconstructs the exact logical input layout for replay. Its explicit dependency lock gives the source, size and SHA-256 digest of every separately acquired input. The Dahmen–Siksek, PVT and BPS papers and unlicensed upstream code snapshots are acquired from their sources rather than redistributed. Project modifications to pinned upstream source files are reconstructed locally and checked against their exact expected digests. Third-party material retained under its own licence is identified in the companion’s attribution notice. Seven historical executables used only for byte identity checks in the earlier package are omitted; their identities and the affected provenance checks are recorded explicitly. No new execution of those binaries or of the historical Magma jobs is claimed.

The public integrity check is

`python3 -B scripts/verify_companion.py.`

It verifies the exact public file set and logical input mapping. The separate script `scripts/replay_companion.py`, supplied with explicit software and input paths, runs the full declared suite from new output directories. The replacement was replayed from a clean extraction with network access and access to earlier proof workspaces blocked: all 13 sector and interface records, all 65 prior jobs, and all 16 rank-local checks passed. The latter consume five freshly generated inputs. The expanded logical inputs and separately acquired private inputs are authenticated before and after execution. The inherited internal identifier `verified-2026-09-14.1` is retained only for verifier compatibility; the replacement has a newly computed input manifest and the public identity printed above.

The companion guide gives acquisition, reconstruction and replay commands, runtime requirements, dependency indexes, successful logs and retained imported premises. The rank route includes the local-image and fake-kernel stopping proofs in Appendix D, the same-literal-*c* comparison and the  $H_0/H_1$  index statement. Supplementary argument-audit records and the seven earlier Magma executions are identified separately. Integrity checking alone is not an arithmetic replay. The principal correspondence is:

| mathematical<br>assertion                            | provenance                                                          | certificate family                                                                                                     |
|------------------------------------------------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| four exhaustive<br>factor sectors<br>Proposition 3.1 | Dahmen–Siksek;<br>reconstructed<br>Dahmen–Siksek;<br>reconstructed  | signed global-algebra superselection<br>rational-factor composition and its<br>Mordell–Weil, sieve, and local packages |
| Proposition 4.2                                      | Dahmen–Siksek;<br>new intrinsic<br>routing                          | database-free quadratic router and<br>quadratic-factor reconstruction                                                  |
| Proposition 5.1                                      | Dahmen–Siksek;<br>new formal-group<br>certification                 | quartic router, Mordell–Weil basis, sieve, and<br>intrinsic formal-group packages                                      |
| Theorem 6.1 and<br>Appendix A                        | new                                                                 | Fano-cover, canonical-quartic,<br>rational-parameter, and split-prime packages                                         |
| exceptional-field<br>elimination<br>Proposition 6.2  | new, using cited<br>modular theorems<br>new terminal<br>composition | Putz–PVT interface and finite $q = 29$<br>comparison packages<br>irreducible-sector composition package                |
| Theorem 1.1                                          | new global<br>composition                                           | complete four-sector proof graph and terminal<br>certificate                                                           |

The archive distinguishes identity checks, arithmetic replay, and the mathematical implications of the resulting certificates. A checksum establishes identity, not a theorem. The adopted Python, PARI/GP and SageMath programs reconstruct the specified arithmetic layers, with errors and precision failures treated as failures. The companion guide distinguishes imported literature results, authenticated Magma execution records, and independently reconstructed calculations. The global rank and local stopping conclusions use the stated general theorems and verified hypotheses as well as the replayed arithmetic.

## 9 Concluding perspective

The proof is not driven by the size of a putative solution. It represents that solution simultaneously by a descent algebra, local factor labels, divisor classes, finite quotient coordinates, and  $p$ -adic analytic data. Each representation imposes an exact compatibility condition. The global contradiction comes from the empty intersection of those conditions.

This organization is useful beyond the present signature. It separates three tasks that are often entangled: enumerating the possible algebraic states, transporting a state to a curve where a finitely generated group acts, and testing the remaining local states with analytic functions. It also makes the proof modular: a computational component can be replaced by an independent implementation without changing the surrounding argument, while the finite interfaces remain explicit.

## A The plane quartic in the pure septic case

Let  $K_7 = \mathbf{Q}(s)$  with  $s^2 = -7$ . In the monomial order

$$x^4, x^3y, x^3z, x^2y^2, x^2yz, x^2z^2, xy^3, xy^2z, xyz^2, xz^3, y^4, y^3z, y^2z^2, yz^3, z^4,$$

write the coefficient of each monomial as  $(U + sW)/2$ . The pairs  $(U, W)$  for  $C_+$  are recorded below with the monomial repeated on every row, so that each coefficient can be quoted and checked independently of the surrounding order.

Table 1: Exact coefficients of the plane quartic  $C_+$ . The coefficient of  $m$  is  $(U + sW)/2$ .

| monomial $m$ | $U$                         | $W$                          |
|--------------|-----------------------------|------------------------------|
| $x^4$        | 526589143431946875/69798278 | -114868376351859375/69798278 |
| $x^3y$       | -1271586888400500/3172649   | -870445613016000/3172649     |
| $x^3z$       | 72385949750625/34899139     | -402884946883125/34899139    |
| $x^2y^2$     | -666163656476955/34899139   | 264021753009735/34899139     |
| $x^2yz$      | -74662865230275/34899139    | 45051674372775/34899139      |
| $x^2z^2$     | -20453292214425/69798278    | 1113321269025/69798278       |
| $xy^3$       | 77961057423858/174495695    | 1554848756622/34899139       |
| $xy^2z$      | 5974117172919/34899139      | 534487214709/34899139        |
| $xyz^2$      | 251941629933/34899139       | 209684218779/34899139        |
| $xz^3$       | -13800316635/69798278       | 9656077725/69798278          |
| $y^4$        | -11467163171457/8724784750  | -8293918213539/8724784750    |
| $y^3z$       | -192270890133/174495695     | -111061125567/174495695      |
| $y^2z^2$     | -4669897617/31726490        | -23997395583/158632450       |
| $yz^3$       | 4101243048/174495695        | -629073270/34899139          |
| $z^4$        | 2                           | 0                            |

The equation and its three first partial derivatives have no common projective zero over  $K_7$ ; this is checked by exact elimination. As an independent good-reduction certificate, both reductions at the primes above 23 are smooth plane quartics. Thus  $C_+$  is smooth, nonhyperelliptic, and has genus 3. The two reductions and their common zeta numerator are recorded in Appendix C. The five points in Theorem 6.1 are

$$\begin{aligned} P_1 &= [-113 - 139s : 32925 + 11475s : -138600 - 57960s], \\ P_2 &= [1109 - 305s : 4325 - 19625s : 163800 - 37800s], \\ P_3 &= [-1 + s : 50 : 0], \\ P_4 &= [5 + s : 400 : 0], \\ P_5 &= [-157167 - 154941s : -82225175 + 2993875s : 186832800]. \end{aligned}$$

The homogeneous forms  $G_t$  and  $H_t$  have degree 5. For each monomial  $m = x^a y^b z^c$  of degree 5, write

$$[m]G_t = \frac{U_{G,m} + sW_{G,m}}{2}, \quad [m]H_t = \frac{U_{H,m} + sW_{H,m}}{2}.$$

Table 2 gives the four rational components for every monomial; a component is displayed as separate numerator and positive denominator. These data define the parameter completely, with no external coefficient file required.

Table 2: Exact coefficient ledger for the inverse parameter  $t = -G_t/H_t$ . For each monomial and each form, the two successive rows are the rational components  $U$  and  $W$  of the coefficient  $(U + sW)/2$ .

| monomial | component | rational value                                                                                                                                           |
|----------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| $G_t$    |           |                                                                                                                                                          |
| $x^5$    | $U$       | $\frac{8724124647223604676837903272778053868067651476251979316420946130788203125}{1262470913738019638514233667224982865889496755254608725372117056}$     |
| $x^5$    | $W$       | $\frac{104539127651167761178849503246013584932810386714540106341619861276432421875}{34086714670926530239884309015074537379016412391874435585047160512}$  |
| $x^4y$   | $U$       | $\frac{160516210802709714993563463849330968569737125845236059453160939760278125}{516465373801917124846731954773856626954794127149612660379502432}$       |
| $x^4y$   | $W$       | $\frac{-1516029210727323321897909052657899390133429038437597834516845689254996875}{4648188364217254123620587592964709642593147144346513943415521888}$    |
| $x^4z$   | $U$       | $\frac{-3431261016436648118361864834749807161605723145608327043665883055988296875}{536865756067092851278177866987423963719508495172022360464492778064}$  |
| $x^4z$   | $W$       | $\frac{-14405986606016247091870808274044492866667442818955446375070000269331640625}{536865756067092851278177866987423963719508495172022360464492778064}$ |
| $x^3y^2$ | $U$       | $\frac{-156765051716182227120181782804196796183006994771851219874584685103262375}{6391259000798724419978307940326475758565577323476456672196342596}$     |
| $x^3y^2$ | $W$       | $\frac{-2797425838906852452662912431708042341593603601309085420184135364665875}{3195629500399362209989153970163237879282788661738228336098171298}$       |

Table 2 (continued)

| monomial  | component | rational value                                                                                                                                            |
|-----------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| $x^3yz$   | $U$       | $-\frac{2971418364998561881697119480380162120182637893006270782575715726325204375}{805298634100639276917266800481135945579262742758033540696739167096}$   |
| $x^3yz$   | $W$       | $\frac{581935828601695609265701070325316097436069202977166807199027638486990625}{805298634100639276917266800481135945579262742758033540696739167096}$     |
| $x^3z^2$  | $U$       | $-\frac{1101411663992169427898989261604416930376364487448681819065769508346053125}{11274180877408949876841735206735903238109678398612469569754348339344}$ |
| $x^3z^2$  | $W$       | $-\frac{107499314624920129138097362917410399320732876511214060737904168396284375}{3758060292469649958947245068911967746036559466204156523251449446448}$   |
| $x^2y^3$  | $U$       | $\frac{456938855157174881485528038353970608056813975973606988967508412987185}{4260839333865816279985538626884317172377051548984304448130895064}$          |
| $x^2y^3$  | $W$       | $\frac{1661711353625849642579873501624741967233922652605715624836234484805605}{12782518001597448839956615880652951517131154646952913344392685192}$        |
| $x^2y^2z$ | $U$       | $\frac{8474129667022123046619324483315222687658910122135070504455178101736425}{115042662014377039559609542925876563654180391822576220099534166728}$       |
| $x^2y^2z$ | $W$       | $\frac{10719261028668655910991493734095990759234174142059301575470250588670875}{268432878033546425639088933493711981859754247586011180232246389032}$      |

Table 2 (continued)

| monomial  | component | rational value                                                                                                                                             |
|-----------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $x^2yz^2$ | $U$       | $\frac{4788522504016908727437175426249647578416822156275084241334728620014875}{16911271316113424815262602810103854857164517597918704354631522509016}$      |
| $x^2yz^2$ | $W$       | $\frac{5591416257739747140557321332572949584770401809552651792943380516721625}{1879030146234824979473622534455983873018279733102078261625724723224}$       |
| $x^2z^3$  | $U$       | $\frac{-673613683718293366731089090657452594859133231945568100511338992953086875}{4261640371660583053446175908146171424005458434675513497367143672272032}$ |
| $x^2z^3$  | $W$       | $\frac{727154975549102369129278091212463518913101596132005437190890995061885625}{4261640371660583053446175908146171424005458434675513497367143672272032}$  |
| $xy^4$    | $U$       | $\frac{181934888585282020021801499997708888564901966896834717514388222336671}{127825180015974488399566158806529515171311546469529133443926851920}$         |
| $xy^4$    | $W$       | $\frac{-66576524945702734300367034318330080309880903641645008587735355126139}{127825180015974488399566158806529515171311546469529133443926851920}$         |
| $xy^3z$   | $U$       | $\frac{55752557283144284245053396819690398445556264921670780430266851605377}{134216439016773212819544466746855990929877123793005590116123194516}$          |
| $xy^3z$   | $W$       | $\frac{-154753525348486993811829134912142488930864130682803774941942077141595}{402649317050319638458633400240567972789631371379016770348369583548}$        |

Table 2 (continued)

| monomial  | component | rational value                                                                                                                                            |
|-----------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| $xy^2z^2$ | $U$       | $\frac{45466049893188675406917805349188754066362340943964804016941934609315}{1537388301464856801387509346373077714287683417992609486784683864456}$        |
| $xy^2z^2$ | $W$       | $\frac{-90644573939421031699724276084922394905211688255008129426169575178615}{1537388301464856801387509346373077714287683417992609486784683864456}$       |
| $xyz^3$   | $U$       | $\frac{86176315804162437100883836603358200621380221474103257474111818725307525}{6392460557490874580169263862219257136008187652013270246050715508408048}$  |
| $xyz^3$   | $W$       | $\frac{-14901625678223850735415766370606467751263679923368054621027598776871975}{6392460557490874580169263862219257136008187652013270246050715508408048}$ |
| $xz^4$    | $U$       | 0                                                                                                                                                         |
| $xz^4$    | $W$       | 0                                                                                                                                                         |
| $y^5$     | $U$       | $\frac{-50208669906374986341132909015146131126007030919811841716909}{18313514842869689191983526987088351258653049645369350436400}$                        |
| $y^5$     | $W$       | $\frac{-118965057882822154213596381597899423461490949136300188722323}{91567574214348445959917634935441756293265248226846752182000}$                       |

Table 2 (continued)

| monomial  | component | rational value                                                                                                                         |
|-----------|-----------|----------------------------------------------------------------------------------------------------------------------------------------|
| $y^4 z$   | $U$       | $-\frac{317869767610131557388806079392661467805959663406188675965981}{164821633585827202727851742883795161327877446808324153927600}$   |
| $y^4 z$   | $W$       | $-\frac{20431957573101268107747459729242315992574863384255199500651}{32964326717165440545570348576759032265575489361664830785520}$     |
| $y^3 z^2$ | $U$       | $-\frac{90218411306089523014211143406416603450262335284699082090763}{346125430530237125728488660055969838788542638297480723247960}$    |
| $y^3 z^2$ | $W$       | $-\frac{6384772760573948911903427720751451666967418352649934952949}{69225086106047425145697732011193967757708527659496144649592}$      |
| $y^2 z^3$ | $U$       | $\frac{34563607476891939880909708193414566015508357013210019043845}{13083541274042963352536871350115659906206911727644771338772888}$   |
| $y^2 z^3$ | $W$       | $-\frac{150431184828356847118153754501573195221720737251858300004385}{13083541274042963352536871350115659906206911727644771338772888}$ |
| $yz^4$    | $U$       | 0                                                                                                                                      |
| $yz^4$    | $W$       | 0                                                                                                                                      |
| $z^5$     | $U$       | 0                                                                                                                                      |
| $z^5$     | $W$       | 0                                                                                                                                      |

Table 2 (continued)

| monomial | component | rational value                                                                                                                                       |
|----------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| $H_t$    |           |                                                                                                                                                      |
| $x^5$    | $U$       | $\frac{417540987588276327046139642388996781228921135238823876465270116076953125}{55097060351713141551537137416607549642806701064452212691347781}$    |
| $x^5$    | $W$       | $\frac{271400501016398695076663568050301774206517949977640898218621509239453125}{55097060351713141551537137416607549642806701064452212691347781}$    |
| $x^4y$   | $U$       | $\frac{40800518279237110617651643358322019052974596058651349157620400597734375}{40070589346700466582936099939350945194768509865056154684616568}$     |
| $x^4y$   | $W$       | $\frac{-10475518577744892148533246091047638248959642317009125512752949147203125}{40070589346700466582936099939350945194768509865056154684616568}$    |
| $x^4z$   | $U$       | $\frac{295466106564310819299307055245131237369729837129047525997756150038046875}{13884459208631711670987358628985102509987288668241957598219640812}$ |
| $x^4z$   | $W$       | $\frac{-97401726931434571033238656526295780061095710676931455291585333023515625}{13884459208631711670987358628985102509987288668241957598219640812}$ |
| $x^3y^2$ | $U$       | $\frac{-2408541684482196326060738206798781548127364231879824910739247944739375}{73462747135617522068716183222143399523742268085936283588463708}$     |
| $x^3y^2$ | $W$       | $\frac{-1401617439695752490670177104164771958374512006836961227347342807905625}{220388241406852566206148549666430198571226804257808850765391124}$    |

Table 2 (continued)

| monomial  | component | rational value                                                                                                                                          |
|-----------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| $x^3yz$   | $U$       | $-\frac{88727646196045391066194147279337983419623095543345635287685584174321875}{13884459208631711670987358628985102509987288668241957598219640812}$    |
| $x^3yz$   | $W$       | $-\frac{294670882430653452222330070241389024327059653868933829299253113096875}{661164724220557698618445648999290595713680412773426552296173372}$        |
| $x^3z^2$  | $U$       | $-\frac{143378643984727743099590687262597995420899449430994725195546663512734375}{1166294573525063780362938124834748610838932248132324438250449828208}$ |
| $x^3z^2$  | $W$       | $-\frac{18365708170095511831980056438251994847190327220498321285191544780928125}{166613510503580540051848303547821230119847464018903491178635689744}$   |
| $x^2y^3$  | $U$       | $\frac{8623734371967307823196434999403662938951554222083721534345295726825}{110194120703426283103074274833215099285613402128904425382695562}$           |
| $x^2y^3$  | $W$       | $\frac{18881132466481791978253130146174121549567643122623579546749820507925}{110194120703426283103074274833215099285613402128904425382695562}$          |
| $x^2y^2z$ | $U$       | $\frac{28562333150244501488893362369966137889574552762163770883392632257375}{661164724220557698618445648999290595713680412773426552296173372}$          |
| $x^2y^2z$ | $W$       | $\frac{127302955155020767473662087940112338634670901473338060096039562505625}{1983494172661673095855336946997871787141041238320279656888520116}$        |

Table 2 (continued)

| monomial  | component | rational value                                                                                                                                          |
|-----------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| $x^2yz^2$ | $U$       | $\frac{111047168050090146883232536414177868796628159548247958136634257244625}{583147286762531890181469062417374305419466124066162219125224914104}$      |
| $x^2yz^2$ | $W$       | $\frac{808931698949954129585743924035840186651521792034283998632202423247375}{194382428920843963393823020805791435139822041355387406375074971368}$      |
| $x^2z^3$  | $U$       | $\frac{56671479059928976818861058588328919604601251570358478252116683982508125}{440859348792474108977190611187534974897116389794018637658670035062624}$ |
| $x^2z^3$  | $W$       | $\frac{71221107904627237798651209022651158397866104168849046635811093612975625}{440859348792474108977190611187534974897116389794018637658670035062624}$ |
| $xy^4$    | $U$       | $\frac{15345599112192921388150576223890339236281253816603221545634911025}{18365686783904380517179045805535849880935567021484070897115927}$              |
| $xy^4$    | $W$       | $\frac{-32203376966478498296624374226292138386494075379549015420791430453}{55097060351713141551537137416607549642806701064452212691347781}$             |
| $xy^3z$   | $U$       | $\frac{3144967508609794715796497440988774261265066246157112647816548077300}{3471114802157927917746839657246275627496822167060489399554910203}$          |
| $xy^3z$   | $W$       | $\frac{-1516919058667068918402436722904874293272390262093914022875222741380}{3471114802157927917746839657246275627496822167060489399554910203}$         |

Table 2 (continued)

| monomial  | component | rational value                                                                                                                                         |
|-----------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| $xy^2z^2$ | $U$       | $\frac{1438698269678151371696913059361694896892235546906064110725007289965}{6626673713210589661153057527470162561584842318933661580968464933}$         |
| $xy^2z^2$ | $W$       | $\frac{-80096480704524798441566620354536123544154597802973268379738357665}{946667673315798523021865361067166080226406045561951654424066419}$           |
| $xyz^3$   | $U$       | $\frac{3102145252841255952780604187946926413931338657791125868394786111962775}{220429674396237054488595305593767487448558194897009318829335017531312}$ |
| $xyz^3$   | $W$       | $\frac{63124143098168155026060265986978624933421514365737963701974316645275}{220429674396237054488595305593767487448558194897009318829335017531312}$   |
| $xz^4$    | $U$       | 0                                                                                                                                                      |
| $xz^4$    | $W$       | 0                                                                                                                                                      |
| $y^5$     | $U$       | $\frac{186871683168118003488935611555116889880084010}{32767303919729427758399266748957493087161279}$                                                   |
| $y^5$     | $W$       | $\frac{25444070085364078630942054472628302788437558}{32767303919729427758399266748957493087161279}$                                                    |

Table 2 (continued)

| monomial | component | rational value |
|----------|-----------|----------------|
| $y^4z$   | $U$       | 2              |
| $y^4z$   | $W$       | 0              |
| $y^3z^2$ | $U$       | 0              |
| $y^3z^2$ | $W$       | 0              |
| $y^2z^3$ | $U$       | 0              |
| $y^2z^3$ | $W$       | 0              |
| $yz^4$   | $U$       | 0              |
| $yz^4$   | $W$       | 0              |
| $z^5$    | $U$       | 0              |
| $z^5$    | $W$       | 0              |

The exact canonical-map identity on  $C_+$  is

$$G_t + tH_t = 0, \quad (41)$$

and therefore  $t = -G_t/H_t$  wherever  $H_t \neq 0$ . This rational function extends uniquely to a morphism  $C_+ \rightarrow \mathbf{P}^1$ , since  $C_+$  is smooth and projective. At a common zero of  $G_t, H_t$ , the value is obtained after cancelling their common vanishing in a local parameter; it cannot be read from the displayed projective pair alone. The finite sieve retains such common-zero reductions conservatively. The certificate verifies (41); the five reference points are not common zeros, and direct substitution gives  $t(P_1) = 0$ ,  $t(P_2) = 1$ , and  $t(P_3) = t(P_4) = t(P_5) = \infty$ . Local expansions on the smooth quartic give

$$\text{ord}_{P_1}(t) = 5, \quad \text{ord}_{P_2}(t - 1) = 3, \quad \text{ord}_{P_3}(1/t) = \text{ord}_{P_4}(1/t) = 7, \quad \text{ord}_{P_5}(1/t) = 1.$$

In particular,  $P_5$  is an unramified point above the branch value  $\infty$ .

## B Integral coordinates for the genus-two formal group

On the divisor-pair chart, write  $\alpha_1 = x_1 + x_2$  and  $\alpha_2 = x_1x_2$ . The unnormalized rational coordinate functions in Flynn's projective embedding include

$$a_{15} = 1, \quad a_{14} = \alpha_1, \quad a_{13} = \alpha_2, \quad a_{12} = \alpha_1^2 - 2\alpha_2, \quad a_{11} = \alpha_1\alpha_2, \quad a_{10} = \alpha_2^2.$$

These formulas use the representative with  $a_{15} = 1$ ; they do not normalize  $a_0$  to 1. Near the projective origin, the normalized coordinates are instead  $s_i = a_i/a_0$ . In particular,  $a_0a_{15} = a_5^2$ , so  $s_{15} = s_5^2$  and  $s_{15}$  vanishes at the origin.

The computations use the following integral coordinate block:

$$b_i = a_i \quad (0 \leq i \leq 11), \quad b_{12} = a_{13}, \quad b_{13} = a_{14}, \quad b_{14} = a_{15}, \quad b_{15} = a_{12} - 2a_{13}.$$

Its inverse is  $a_{12} = b_{15} + 2b_{12}$ , with all other displayed coordinates recovered directly. Hence this is an integral unimodular change of basis. Membership in the depth-two origin neighbourhood is therefore equivalent in the  $a$ - and  $b$ -coordinates. The valuation rows in (23) are stated in the  $b$ -coordinates.

## C The split-prime data at 23 for the rational-parameter theorem

The prime 23 splits in  $K_7$  as the two embeddings  $s \mapsto 4$  and  $s \mapsto 19$ . Both reductions of  $C_+$  have 20 rational points. Their common local numerator of the zeta function is

$$1 - 4T + 4T^2 + 56T^3 + 92T^4 - 2116T^5 + 12167T^6,$$

so

$$\#J_+(\mathbf{F}_{23}) = 10200, \quad 23 \nmid 10200.$$

The ordered reductions of the five points are

|       | $s = 4$     | $s = 19$    |
|-------|-------------|-------------|
| $P_1$ | (1, 21, 1)  | (1, 11, 15) |
| $P_2$ | (1, 0, 22)  | (1, 8, 14)  |
| $P_3$ | (1, 9, 0)   | (1, 13, 0)  |
| $P_4$ | (1, 1, 0)   | (1, 9, 0)   |
| $P_5$ | (1, 11, 21) | (1, 6, 9).  |

For the regular differential basis  $\eta_x, \eta_y, \eta_z$ , the corresponding evaluation vectors are

|       | $s = 4$     | $s = 19$      |
|-------|-------------|---------------|
| $P_1$ | (3, 17, 3)  | (11, 6, 4)    |
| $P_2$ | (4, 0, 19)  | (5, 17, 1)    |
| $P_3$ | (4, 13, 0)  | (2, 3, 0)     |
| $P_4$ | (12, 12, 0) | (22, 14, 0)   |
| $P_5$ | (7, 8, 9)   | (10, 14, 21). |

These data independently reproduce the tangent determinants in (32) once the two annihilator rows are applied.

## D The complete rank-four descent and its downstream use

Let  $K = K_7 = \mathbf{Q}(s)$ , where  $s^2 = -7$ , and let  $J = \text{Jac}(C_+)$  for the smooth plane quartic of Appendix A. Write  $D_i = [P_i - P_1]$  and

$$H_0 = \langle D_2, D_3, D_4, D_5 \rangle, \quad H_1 = \langle D_2, E, D_4, D_5 \rangle.$$

We prove that  $J(K)$  has rank four, that  $H_0$  has odd index, and that  $[J(K) : H_1]$  is coprime to 400. These are the group-theoretic inputs to the sieve and logarithmic argument in Theorem 6.1. The relation (31) gives  $H_0 \subseteq H_1$ ; the 2-saturation statement concerns  $H_0$ , and the 5-saturation statement concerns  $H_1$ .

### D.1 The descent maps and the local completeness criterion

Let  $\Delta$  be the Galois set of the 28 bitangents of  $C_+$ , and let  $L = (\overline{K}^\Delta)^{\text{Gal}(\overline{K}/K)}$  be its degree-28 étale algebra of Galois-equivariant functions. Evaluation of the normalized bitangent lines on a degree-zero divisor  $z$  gives  $f(z) \in L^\times$ , whenever the divisor avoids their zeroes and poles. For  $F = K$  or a completion  $K_v$ , put

$$\mathcal{H}(F) = (L \otimes_K F)^\times / (F^\times (L \otimes_K F)^{\times 2}).$$

The evaluation construction induces the fake descent map

$$f_F : J(F)/2J(F) \longrightarrow \mathcal{H}(F).$$

Thus a divisor class is in the *fake kernel* precisely when its evaluation can be written  $f(z) = a\lambda^2$ , with  $a \in F^\times$  and  $\lambda \in (L \otimes_K F)^\times$ .

For clarity, the finite modules in the comparison with true descent are as follows. Let  $E_\Delta \subset \mathbf{F}_2^\Delta$  be the subspace of even subsets, and let  $R_\Delta \subset E_\Delta$  be the span of the 315 syzygetic tetrads. Write  $E_\Delta^\vee, R_\Delta^\vee$  for their duals and  $q : E_\Delta^\vee \rightarrow R_\Delta^\vee$  for restriction. The exact bitangent and contact identities identify

$$\ker q \simeq J[2], \quad \dim E_\Delta = 27, \quad \dim R_\Delta = 21,$$

as Galois modules, by [9, Corollary 12.5]. The rational even theta characteristic and the seven blocks in the exact field tower put the actual Galois action inside  $\text{AGL}_3(\mathbf{F}_2)$ , acting transitively on  $\Delta$ . Equality with the full affine group is not needed. The rational point  $P_1$  supplies the degree-one divisor hypotheses for the global and local comparison results of [9, Lemma 10.2 and Appendix A].

If  $M$  is one of these finite Galois modules,  $M(F)$  denotes its  $F$ -rational elements. Define the correction quotient

$$\mathcal{C}(F) = R_\Delta^\vee(F)/q(E_\Delta^\vee(F)).$$

The tetrad incidence map  $\tau_*$  multiplies the four corresponding bitangent values. The normalized contact conics give an evaluation  $r(z)$  satisfying the exact identity

$$r(z)^2 = \tau_*(f(z)).$$

Consequently, if  $f(z) = a\lambda^2$ , the correction of  $z$  is represented by

$$\frac{r(z)}{a^2\tau_*(\lambda)}. \quad (42)$$

The scalar exponent is two because each tetrad has four entries. Formula (42) is the identity of [9, Lemma A.24(a)]. Its entries lie in  $\mu_2$  before any local approximation is made. A nonzero value under a character of  $\mathcal{C}(F)$  therefore proves that the class of  $z$  is nonzero in  $\ker f_F$ .

**Lemma D.1** (Local completeness). *Let  $d_v = \dim_{\mathbf{F}_2} J(K_v)/2J(K_v)$ . Suppose actual local divisor classes give  $r_v$  independent fake images and  $k_v$  independent classes in the fake kernel. If  $r_v + k_v = d_v$ , these images span the full fake image and the fake kernel has dimension  $k_v$ .*

*Proof.* Apply rank–nullity to  $f_{K_v}$ . The two exhibited subspaces give lower bounds whose sum already equals the dimension of its domain. For a finite place of odd residue characteristic,  $d_v = \dim J[2](K_v)$ ; for  $v \mid 2$ ,  $d_v = 3[K_v : \mathbf{Q}_2] + \dim J[2](K_v)$ , by [9, Remark 11.6]. This is the stopping criterion of [9, Lemma A.24(b)].  $\square$

## D.2 A containing space for the global fake image

Let  $S$  consist of the complex place, both dyadic places, the places above 3, 5, 7, and the bad place  $(29, s - 15)$ . The primitive quartic is smooth outside  $S$ , and the bitangent algebra is unramified there. The base field  $K$  has class number one. The containing direction of [9, Propositions 7.2–7.3 and Theorem 10.9(a)] therefore puts every relevant fake Selmer class in the image of  $L(S, 2)$ , the squareclasses with even valuation outside the primes above  $S$ .

The unconditional class-group calculation gives  $\text{Cl}(L)[2] = 0$ . The 38 finite primes of  $L$  above  $S$  are principal, and the unit calculation gives 28 independent ordinary-unit squareclasses, including  $-1$ . The exact valuation sequence for  $S$ -unit squareclasses shows that these 28 classes and the 38 prime generators form a basis of the 66-dimensional space

$$V_0 = L(S, 2) \subset L^\times / L^{\times 2}.$$

Thus every fake image of a true 2-Selmer class has a representative in  $V_0$ . For a selected place  $v$ , impose on  $V_0$  the condition that its restriction modulo the diagonal  $K_v^\times$  belong to  $\text{im } f_{K_v}$ .

The local image dimensions used here are

| place                                  | $d_v$ | $\dim \text{im } f_{K_v}$ | $w = (1 + s)/2.$ |
|----------------------------------------|-------|---------------------------|------------------|
| four completely split bitangent places | 6     | 6                         |                  |
| $v_{1051} = (1051, s - 815)$           | 4     | 4                         |                  |
| $v \mid 3$                             | 2     | 2                         |                  |
| $v \mid 5$                             | 3     | 3                         |                  |
| $v \mid 7$                             | 2     | 1                         |                  |
| $v \mid 2, w \equiv 1 \pmod{2}$        | 4     | 4                         |                  |
| $v \mid 2, w \equiv 0 \pmod{2}$        | 4     | 3                         |                  |

At the four good places above 19163, 23431, 29059, 38189, the bitangent algebra splits completely. Although 1051 also splits in  $K/\mathbf{Q}$ , at the selected good place  $v_{1051} = (1051, s - 815)$  the bitangent algebra has residue factor degrees  $1^8 2^{10}$ . Thus  $L \otimes_K K_{v_{1051}} \simeq K_{v_{1051}}^8 \times E_{1051}^{10}$ , where  $E_{1051}/K_{v_{1051}}$  is the unramified quadratic extension. Its Frobenius action gives  $\dim_{\mathbf{F}_2} J[2](K_{v_{1051}}) = 4$ . For the first four rows, independent point images attain the local two-torsion bound, so Lemma D.1

applies with  $k_v = 0$ . The 7-adic and dyadic arguments below justify the remaining rows, including their kernel contributions.

Let  $V \subset V_0$  be the intersection of these conditions. The restriction of each of the 66 generators is computed in every factor of the actual local bitangent algebra. Exact linear algebra gives

$$66 \longrightarrow 23 \longrightarrow 22 \longrightarrow 20 \longrightarrow 19 \longrightarrow 18 \longrightarrow 15 \longrightarrow 12. \quad (43)$$

The stages are the four completely split bitangent places together,  $v_{1051}$ , the places above 3, 5, 7, and the dyadic places with  $w \equiv 1, 0$ , in that order. The other prime conditions and the norm condition in the ancillary descent calculation are redundant for this intersection.

The diagonal subspace

$$D = V \cap \text{im}(K^\times / K^{\times 2} \longrightarrow L^\times / L^{\times 2})$$

has dimension seven. Set  $Q = V/D$  and let  $B$  be the fake image of  $H_0$ . The coordinate calculation gives

$$Q = B \oplus \langle c \rangle, \quad \dim B = 4, \quad \dim Q = 5. \quad (44)$$

Here  $c$  denotes the fixed complementary squareclass in the descent data; we use the same representative in  $L^\times$  throughout the comparison below. The calculation verifies that the diagonal, the four point images, and this representative satisfy every local condition, have ranks 7, 11, 12 successively, and span  $V$ . Thus (44) is an equality of the specified spaces. In particular, the fake image of every true Selmer class lies in  $Q$ . We do not need every element of  $Q$  to be a Selmer class.

### D.3 Completeness at seven

Let  $v \mid 7$ . The actual decomposition data give  $d_v = 2$ . For the divisor  $z_7 = P_4 - P_1$ , exact square tests in all seven factors of  $L \otimes_K K_v$  give

$$f(z_7) = 3\lambda^2.$$

This puts its class in the full fake kernel. To prove it is a nonzero class, consider the specified completion  $A$  of the degree-42 tetrad algebra. It has  $(e, f) = (4, 1)$  over  $K_v$ , and the incidence algebra over  $A$  is a product of two unramified quadratic extensions. Evaluation on this tetrad component defines a character of  $\mathcal{C}(K_v)$ : this is checked for every affine-group candidate compatible with the measured decomposition data. Some of those characters vanish, so their factor profiles alone do not prove nonvanishing.

In the two quadratic incidence fields, remove the even valuations from  $f(z_7)/3$  and take square roots of the resulting units. Their residue roots over  $\mathbf{F}_{49}$  are simple and hence lift by Hensel's lemma. The two normalized norm residues are one; the corresponding normalized conic quotient has residue six. Formula (42) therefore gives

$$\frac{r(z_7)}{3^2 \tau_*(\lambda)} = -1$$

on the selected target. The equality is exact: the contact identity puts the quotient in  $\mu_2$ , and reduction in residue characteristic seven distinguishes 1 from  $-1$ . Thus  $\ker f_{K_v}$  is nonzero. For comparison, the distinct divisor  $P_2 - P_1$ , also with diagonal scalar 3, gives  $+1$ . Changing either quadratic square-root sign does not change its norm.

There is also a nonzero fake image. Let  $K'_v/K_v$  be the unramified quadratic extension. The local-point calculation produces  $P \in C_+(K'_v)$  such that the degree-zero divisor

$$\text{Norm}_{K'_v/K_v}(P) - 2P_1$$

has nonzero fake image. The coordinate change from its local model to the printed quartic is  $Z_{\text{original}} = -7Z_{\text{local}}$ . A strong Hensel inequality proves existence of the point, and all seven line-value squareclasses are unchanged by the correction. Its norm-line vector, modulo the two diagonal classes, spans the one-dimensional allowed image used in (43). We have now exhibited an image of dimension at least one and a kernel of dimension at least one in a domain of dimension two. Lemma D.1 proves that both dimensions are exactly one.

**Precision for the 7-adic calculation.** The target is represented by an integral Eisenstein polynomial of degree eight over  $\mathbf{Q}_7$ . Strong Hensel bounds and Krasner’s lemma identify the computed generator with the exact target root; Gauss bounds propagate its error through the rational line, incidence and conic coefficients, including their denominators. The two incidence factors lift from coprime factors. A uniformizer change in one factor gives its irreducible residual quadratic. At working precision 100 over  $\mathbf{Q}_7$ , the resulting relative-error bound is at least 254 in  $A$ , so the residue used above is unchanged. The computed sign separations at working precisions 100 and 120 are 255 and 511. These inequalities justify passage from approximations to the exact sign, rather than relying on agreement of two numerical outputs.

#### D.4 Completeness at the dyadic places

At both dyadic places  $d_v = 4$ . At the place with  $w \equiv 1 \pmod{2}$ , the known point differences together with the Hensel lift of  $[2 : 46 : 1]$  give four independent fake images. This already attains  $d_v$ , so the image is complete and its fake kernel is zero.

Now work at the place with  $w \equiv 0 \pmod{2}$ , identified with  $\mathbf{Q}_2$ . The four known differences have fake-image rank two. We describe the additional points and kernel divisors explicitly. In the chart

$$[X : Y : Z] = [13 + 32e : 1 : 16d],$$

let  $C_0 = [13 : 1 : 16d_0]$ , where  $d_0 \equiv 0 \pmod{2}$  is the Hensel root of the quartic equation. The normalized equation reduces to  $d(d+1)^3$ , so this root is unique in its residue disk. Its fake image is independent of the known images, giving rank at least three.

In the chart  $[13 + 32e : 1 : 16(1 + 2g)]$ , the normalized reduction is

$$e^4 + e^3 + g^2e^2 + (g^2 + g)e + g^3 + g^2 + g = 0.$$

Its  $g$ -derivative at  $(0,0)$  and its  $e$ -derivative at  $(1,0)$  are units. Hence the following points are well defined by their coordinates and congruences:

$$\begin{aligned} P'_2 &= [77 : 1 : 16(1 + 2g_2)], & g_2 &\equiv 0 \pmod{2}, \\ P'_5 &= [13 + 32e_5 : 1 : 80], & e_5 &\equiv 1 \pmod{2}. \end{aligned}$$

Here  $g_2$  is obtained by fixing  $e = 2$ , and  $e_5$  by fixing  $g = 2$ . Define the degree-zero divisors

$$z_2 = P'_2 + P_2 + C_0 - 3P_1, \quad z_5 = P'_5 + P_2 - 2P_1.$$

Exact square tests in all three factors of the local bitangent algebra give

$$f(z_2) = -10\lambda_2^2, \quad f(z_5) = 2\lambda_5^2.$$

Thus these are actual fake-kernel divisors. The line-value error bounds put the ratio of each true value to its tested approximation in  $1 + \mathfrak{p}^{2\text{ord}_{\mathfrak{p}}(2)+1}$ , whose elements are squares by Hensel’s lemma.

The degree-42 target splits here into factors of degrees 3, 3, 6, 6, 24. Evaluation on the two sextic factors gives two independent characters of the two-dimensional quotient  $\mathcal{C}(K_v)$ . Both characters annihilate  $q(E_{\Delta}^{\vee}(K_v))$  for all 28 compatible affine-group candidates. With  $-1$

represented by the bit 1, both  $z_2$  and  $z_5$  have correction values  $(0, 1)$  under these characters. In particular the fake kernel is nonzero. The rank-three image and this kernel class fill the four-dimensional local Kummer group; Lemma D.1 proves that the full fake image has dimension three. This proves the last two cuts in (43).

**Precision for the dyadic correction.** Integral local bases, ramification and residue polynomials identify the two sextic fields and their quadratic incidence factors. Strong Hensel and Krasner bounds apply to the original target polynomial, and the factor-lifting error is propagated through the evaluations. In the strengthened calculation for  $z_5$ , the unit square-root residual and derivative valuations are  $(40, 4)$  in a ramified quadratic factor and  $(20, 2)$  in an unramified one. The corresponding exact-root error bounds are 36 and 18; after taking norms the relative error in the target is at least 18. The final sign-separation pairs are  $(19, 2)$  and  $(2, 18)$ ; in each sextic target field  $A$ ,  $\text{ord}_A(2) = 2$ . The exact contact identity therefore distinguishes the signs despite residue characteristic two. Only a valid square-root witness is required: changing its sign has no effect on a quadratic norm, so uniqueness of the finite root search is unnecessary.

## D.5 Excluding the complementary class

We next compare the containing space  $Q$  with the true 2-Selmer group. In fact,  $J(K)[2] = 0$  follows already from transitivity on  $\Delta$ . Identify  $E_\Delta^\vee$  with  $\mathbf{F}_2^\Delta / \langle \mathbf{1} \rangle$  using the standard dot product. The bitangent realization gives  $J[2] \simeq R_\Delta^\perp / \langle \mathbf{1} \rangle$ . The weight enumerator of  $R_\Delta^\perp$  is

$$1 + 63T^{12} + 63T^{16} + T^{28};$$

it is obtained by taking the orthogonal complement of the 315 tetrad incidence vectors, which have rank 21, and enumerating its 128 vectors. Each nonzero quotient class therefore has two complementary representatives of weights 12 and 16. If such a class were Galois invariant, its unique weight-12 representative would have Galois-invariant support, a nonempty proper subset of the transitive set  $\Delta$ . This is impossible. Separately, the finite Galois-module calculation gives  $\dim \mathcal{C}(K) = 1$  for every transitive affine subgroup compatible with the actual bitangent action.

The fixed representative  $c$  in (44) has two coherent global lifts under the BPS comparison map on  $H^1(K, J[2])$ . Here coherence includes the relations required by [9, Lemma A.13(b)], rather than only the existence of square roots of  $\tau_*(c)$ . More explicitly, the degree-42 and degree-56 target orbits give 98 incidence vectors spanning  $R_\Delta$ . Exact target norm roots satisfy the 77 independent relations among those vectors. Their relation quotients lie in  $\mu_2$ , so their exact signs are determined by reduction at the odd split prime 733177, where all denominators are units. This defines the lift on all 315 tetrads and respects the Galois action. The two compatible whole-orbit sign choices, denoted 00 and 11, differ by the nonzero element  $\rho \in \mathcal{C}(K)$ . Since this quotient has dimension one, these are all the lifts.

At  $v \mid 5$  the fake map is injective: the three independent images already attain  $d_v = 3$ . Hence the local correction quotient used to test whether a cohomology class is Kummer is  $\mathcal{C}(K_v)$  itself; there is no local fake-kernel contribution to quotient out. Its dimension is two. For the actual divisor  $z = P_2 + P_4 - 2P_1$  and  $a = (s - 1)/2$ , the same global representative satisfies

$$f(z) = ca\lambda^2$$

in all eleven components of  $L \otimes_K K_v$ . If  $u^2 = \tau_*(c)$  is a coherent target root, the local obstruction is represented by

$$\frac{u a^2 \tau_*(\lambda)}{r(z)}.$$

Its square is one by the exact contact and norm identities. The physical target characters give

$$\omega_{00} = (1, 0), \quad \omega_{11} = (0, 1), \quad \kappa_5(\rho) = (1, 1).$$

Here  $\kappa_5$  is localization on the correction quotient; its value is the difference of the two obstructions, not either obstruction itself. The functional  $(u, v) \mapsto u + v$  vanishes on that difference and is nonzero on both obstructions. Thus neither lift of  $c$  is locally Kummer at 5.

By [9, Theorem 10.14 and Lemma A.26], the kernel of the true-to-fake Selmer comparison is the kernel of localization from  $\mathcal{C}(K)$  to the local correction quotients. Its restriction at 5 is already injective, since  $\kappa_5(\rho) \neq 0$ . The Selmer comparison kernel is consequently zero.

**Arithmetic identification of the 5-adic obstructions.** The exact coordinate representative of  $c$ , the target norm roots, and the local source equation above are compared directly. In the selected quadratic incidence fields, taking norms of  $\lambda^2$  and prescribing the residue of the norm of  $\lambda$  determines the required square root. The three physical character values are  $(-1, +1, -1)$ ; the first and third characters agree, and the first two are independent. Reversing the coherent root gives  $(+1, -1, +1)$ . The relative root-error bounds are at least 21, so these residues determine the exact  $\mu_2$  values. This is the arithmetic underlying the two displayed obstruction vectors.

## D.6 Rank, subgroup indices and the logarithmic annihilator

**Proposition D.2.** *The true 2-Selmer group and the Mordell–Weil group satisfy*

$$\dim_{\mathbf{F}_2} \text{Sel}^2(J/K) = \text{rank } J(K) = 4, \quad \text{Sha}(J/K)[2] = 0.$$

*Proof.* Every true Selmer class has fake image in  $Q = B \oplus \langle c \rangle$ . If its image were  $c + b$ , choose a point of  $H_0$  with fake image  $b$ . Subtracting its global Kummer class would give an everywhere locally Kummer lift of  $c$ , contradicting the obstruction at 5. Thus the true Selmer image lies in  $B$ . The comparison is injective, so the Selmer dimension is at most four. The independent point lower bound is four. Since  $J(K)[2] = 0$ , the Kummer exact sequence

$$0 \longrightarrow J(K)/2J(K) \longrightarrow \text{Sel}^2(J/K) \longrightarrow \text{Sha}(J/K)[2] \longrightarrow 0$$

proves all three assertions. □

The verified 2-saturation, equivalently the independence of the four generators modulo 2 together with this rank calculation, gives  $[J(K) : H_0]$  odd. The relation (31) gives  $H_0 \subseteq H_1$ , so  $H_1$  also has odd index. Its separately verified 5-saturation then gives

$$\gcd([J(K) : H_1], 400) = 1.$$

Multiplication by 400 is therefore an automorphism of the finite group  $J(K)/H_1$ , which proves that  $H_1 \rightarrow J(K)/400J(K)$  is surjective. This is precisely the coefficient coverage required by the finite sieve; saturation at other primes is not used.

Finally, let a characteristic-zero logarithmic functional vanish on  $H_0$ . Every point of  $J(K)$  has a nonzero integer multiple in  $H_0$ , and the target of the functional has no integer torsion. The functional therefore vanishes on all of  $J(K)$ . This proves that the logarithmic annihilator used in Theorem 6.1 is global.

**Location of the arithmetic evidence.** The ancillary component guide `PROGRAMS_AND_CERTIFICATES.md` identifies the ordered 66 representatives, all restriction columns, the diagonal and point coordinates, the fixed representative  $c$ , and the exact root and local-divisor witnesses used here. It also locates the full local-field and precision calculations behind the inequalities above. These data establish the computational premises of the argument; checksums and saved status messages serve only to identify and reproduce them.

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